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Antenna assembly and electronic device

Abstract

Provided are an antenna assembly and an electronic device. The antenna assembly includes a first radiator, a second radiator, a first matching module, a first feeding module, a second matching module, and a second feeding module. The first radiator has a first ground end, a first coupling end, and a first feeding point. The second radiator has a second coupling end, a second ground end, and a second feed point. A first coupling gap is defined between the second coupling end and the first coupling end. The first matching module is electrically connected between the first feeding point and the first feeding module. The second matching module is electrically connected between the second feeding point and the second feeding module. The first radiator and the second radiator support multiple resonant modes, where at least one resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/CN2022/086365, filed Apr. 12, 2022, which claims priority to Chinese Patent Application No. 202110582433.2, filed May 26, 2021, the entire disclosures of which are hereby incorporated by reference.

TECHNICAL FIELD

(1) This disclosure relates to the field of communication technology, and in particular, to an antenna assembly and an electronic device.

BACKGROUND

(2) With the development of communication technologies, electronic devices with communication functions are becoming more and more popular, and the requirements for Internet speed are getting higher and higher. Therefore, how to improve the data transmission rate of the electronic device and improve the communication quality of the electronic device becomes a technical problem to be solved.

SUMMARY

(3) In a first aspect, an antenna assembly is provided in the present disclosure. The antenna assembly includes a first radiator, a second radiator, a first matching module, a first feeding module, a second matching module, and a second feeding module. The first radiator has a first ground end, a first coupling end, and a first feeding point between the first ground end and the first coupling end. The second radiator has a second coupling end, a second ground end, and a second feeding point between the second coupling end and the second ground end. A first coupling gap is defined between the second coupling end and the first coupling end. The first matching module is electrically connected between the first feeding point and the first feeding module, and the first ground end is electrically connected to a first reference ground. The second matching module is electrically connected between the second feeding point and the second feeding module, and the second ground end is electrically connected to a second reference ground. The first radiator and the second radiator support, under excitation of the first feeding module and the second feeding module, multiple resonant modes, where at least one resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by excitation current of the first feeding module resonating on the second radiator.

(4) In a second aspect, an electronic device is provided in the present disclosure. The electronic device includes a housing and at least one above antenna assembly, and radiators of the at least one antenna assembly are integrated into the housing, or are disposed on a surface of the housing, or are disposed in space enclosed by the housing.

(5) Other features and aspects of the disclosed features will become apparent from the following

detailed description, taken in conjunction with the accompanying drawings, which illustrate, by way of example, the features in accordance with implementations of the disclosure. The summary is not intended to limit the scope of any implementations described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In order to describe technical solutions in embodiments of the disclosure more clearly, the following will give a brief introduction to the accompanying drawings required for describing embodiments. Apparently, the accompanying drawings hereinafter described are merely some embodiments of the disclosure. Based on these drawings, those of ordinary skill in the art can also obtain other drawings without creative effort.
- (2) FIG. 1 is a schematic structural diagram of an electronic device provided in embodiments of the present disclosure.
- (3) FIG. 2 is a schematic structural exploded view of the electronic device illustrated in FIG. 1.
- (4) FIG. 3 is a schematic structural diagram of a first type of antenna assembly of the electronic device illustrated in FIG. 2.
- (5) FIG. 4 is a schematic graph illustrating S-parameters of the antenna assembly illustrated in FIG. 3.
- (6) FIG. 5 is a schematic diagram illustrating the distribution of a first-resonant-current density of the antenna assembly illustrated in FIG. 3.
- (7) FIG. 6 is a schematic diagram illustrating the distribution of a second-resonant-current density of the antenna assembly illustrated in FIG. 3.
- (8) FIG. 7 is a schematic diagram illustrating the distribution of a third-resonant-current density of the antenna assembly illustrated in FIG. 3.
- (9) FIG. 8 is a schematic diagram illustrating the distribution of a fourth-resonant-current density of the antenna assembly illustrated in FIG. 3.
- (10) FIG. 9 is a schematic diagram illustrating the distribution of a fifth-resonant-current density of the antenna assembly illustrated in FIG. 3.
- (11) FIG. 10 is a schematic structural diagram of a first matching module illustrated in FIG. 3.
- (12) FIG. 11 is a schematic structural diagram of a first type of first sub-branch provided in embodiments of the present disclosure.
- (13) FIG. 12 is a schematic structural diagram of a second type of first sub-branch provided in embodiments of the present disclosure.
- (14) FIG. 13 is a schematic structural diagram of a third type of first sub-branch provided in embodiments of the present disclosure.
- (15) FIG. 14 is a schematic structural diagram of a fourth type of first sub-branch provided in embodiments of the present disclosure.
- (16) FIG. 15 is a schematic structural diagram of a fifth type of first sub-branch provided in embodiments of the present disclosure.
- (17) FIG. 16 is a schematic structural diagram of a sixth type of first sub-branch provided in embodiments of the present disclosure.
- (18) FIG. 17 is a schematic structural diagram of a seventh type of first sub-branch provided in embodiments of the present disclosure.
- (19) FIG. 18 is a schematic structural diagram of an eighth type of first sub-branch provided in embodiments of the present disclosure.
- (20) FIG. 19 is a schematic structural diagram of the first type of antenna assembly illustrated in FIG. 3 with an adjustable branch.
- (21) FIG. 20 is a detailed structural schematic diagram of a first matching module in FIG. 3.

(22) FIG. 21 is a schematic graph illustrating comparison between S-parameters of the antenna assembly illustrated in FIG. 3 and S-parameters of the antenna assembly illustrated in FIG. 19.

(23) FIG. 22 is a schematic structural diagram of a second matching module illustrated in FIG. 3.

(24) FIG. 23 is a first schematic structural diagram of an electronic device with the first type of antenna assembly illustrated in FIG. 3 mounted thereon.

(25) FIG. 24 is a structural schematic diagram of the electronic device illustrated in FIG. 3 in landscape mode.

(26) FIG. 25 is a schematic graph illustrating efficiency of the first type of antenna assembly illustrated in FIG. 3.

(27) FIG. 26 is a second schematic structural diagram of the electronic device with the first type of antenna assembly illustrated in FIG. 3 mounted thereon.

(28) FIG. 27 is a third schematic structural diagram of the electronic device with the first type of antenna assembly illustrated in FIG. 3 mounted thereon.

(29) FIG. 28 is a first schematic structural diagram of an electronic device with a second type of antenna assembly mounted thereon provided in embodiments of the present disclosure.

(30) FIG. 29 is a schematic structural diagram of the second type of antenna assembly provided in embodiments of the present disclosure.

(31) FIG. 30 is a second schematic structural diagram of the electronic device with the second type of antenna assembly illustrated in FIG. 29 mounted thereon.

(32) FIG. 31 is a schematic structural diagram illustrating improvement of the second type of the antenna assembly illustrated in FIG. 29.

(33) FIG. 32 is a schematic structural diagram of the second type of antenna assembly illustrated in FIG. 29 integrated with proximity detection.

(34) FIG. 33 is a schematic structural diagram of a third type of antenna assembly provided in embodiments of the present disclosure.

(35) FIG. 34 is a schematic graph illustrating S-parameters of a first antenna unit of the antenna assembly illustrated in FIG. 33.

(36) FIG. 35 is a schematic graph illustrating S-parameters of a second antenna unit of the antenna assembly illustrated in FIG. 33.

(37) FIG. 36 is a schematic graph illustrating S-parameters of a third antenna unit of the antenna assembly illustrated in FIG. 33.

(38) FIG. 37 is a schematic diagram illustrating the distribution of a sixth-resonant-current density of the antenna assembly illustrated in FIG. 33.

(39) FIG. 38 is a schematic diagram illustrating the distribution of a seventh-resonant-current density of the antenna assembly illustrated in FIG. 33.

(40) FIG. 39 is a schematic diagram illustrating the distribution of an eighth-resonant-current density of the antenna assembly illustrated in FIG. 33.

(41) FIG. 40 is a schematic graph illustrating S-parameters of a fourth antenna unit of the antenna assembly illustrated in FIG. 33.

(42) FIG. 41 is a schematic diagram illustrating the distribution of a tenth-resonant-current density of the antenna assembly illustrated in FIG. 33.

(43) FIG. 42 is a schematic diagram illustrating the distribution of an eleventh-resonant-current density of the antenna assembly illustrated in FIG. 33.

(44) FIG. 43 is a schematic diagram illustrating the distribution of a twelfth-resonant-current density of the antenna assembly illustrated in FIG. 33.

(45) FIG. 44 is a schematic structural diagram of an electronic device with the antenna assembly illustrated in FIG. 33 mounted thereon.

(46) FIG. 45 is a schematic graph illustrating S-parameters of the antenna assembly illustrated in FIG. 33.

(47) FIG. 46 is a schematic graph illustrating isolation between antenna units of the antenna

assembly illustrated in FIG. 33.

(48) FIG. 47 is a schematic structural diagram of the third type of antenna assembly with a sharing radiator being used as a proximity sensing electrode provided in embodiments of the present disclosure.

(49) FIG. 48 is a schematic structural diagram of the third type of antenna assembly with a second radiator being used as a proximity sensing electrode provided in embodiments of the present disclosure.

(50) FIG. 49 is a schematic structural diagram of the third type of antenna assembly with a fourth radiator being used as a proximity sensing electrode provided in embodiments of the present disclosure.

(51) FIG. 50 is a schematic structural diagram illustrating the third type of antenna assembly with both a sharing radiator and a fourth radiator being used as a proximity sensing electrode provided in embodiments of the present disclosure.

(52) FIG. 51 is a schematic structural diagram of the third type of antenna assembly with a seventh matching module provided in embodiments of the present disclosure.

(53) FIG. 52 is a structural diagram of a third antenna assembly with a seventh matching module and an eighth matching module provided in embodiments of the present disclosure.

DETAILED DESCRIPTION

(54) In a first aspect, an antenna assembly is provided in the present disclosure. The antenna assembly includes a first radiator, a second radiator, a first matching module, a first feeding module, a second matching module, and a second feeding module. The first radiator has a first ground end, a first coupling end, and a first feeding point between the first ground end and the first coupling end. The second radiator has a second coupling end, a second ground end, and a second feeding point between the second coupling end and the second ground end. A first coupling gap is defined between the second coupling end and the first coupling end. The first matching module is electrically connected between the first feeding point and the first feeding module, and the first ground end is electrically connected to a first reference ground. The second matching module is electrically connected between the second feeding point and the second feeding module, and the second ground end is electrically connected to a second reference ground. The first radiator and the second radiator support, under excitation of the first feeding module and the second feeding module, multiple resonant modes, where at least one resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by excitation current of the first feeding module resonating on the second radiator.

(55) In an implementation, the antenna assembly is configured to support a first resonant mode, a second resonant mode, and a fourth resonant mode. The first resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the first feeding module resonating on the first radiator. The second resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength caused by the excitation current of the first feeding module resonating on the second radiator. The fourth resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength caused by the excitation current of the first feeding module resonating from the first feeding point to the first coupling end.

(56) In an implementation, a resonant frequency of the first resonant mode, a resonant frequency of the second resonant mode, and a resonant frequency of the fourth resonant mode increase sequentially. The first resonant mode supports at least one of a global positioning system (GPS) frequency band, a long-term evolution 4th generation (LTE-4G) middle-high band (MHB) frequency band, or a new radio 5th generation (NR-5G) MHB frequency band. The second resonant mode supports at least one of a wireless fidelity 2.4 GHz (Wi-Fi 2.4G) frequency band, the LTE-4G MHB frequency band, or the NR-5G MHB frequency band. The fourth resonant mode supports at least one of an LTE-4G ultra-high band (UHB) frequency band or an NR-5G UHB frequency band.

(57) In an implementation, the first matching module includes at least one first branch connected in parallel to the ground and/or at least one second branch connected in series between the first

feeding point and the first feeding module, and each of the at least one first branch and the at least one second branch includes at least one of a capacitor or an inductor. The first matching module is configured to tune at least one of a resonant frequency of the first resonant mode, a resonant frequency of the second resonant mode, or a resonant frequency of the fourth resonant mode.

(58) In an implementation, the at least one first branch includes a first sub-branch, a second sub-branch, and a third sub-branch, and the at least one second branch includes a fourth sub-branch and a fifth sub-branch. One end of the first sub-branch is electrically connected to the first feeding point, the other end of the first sub-branch is electrically connected to a third reference ground, and the first sub-branch includes a capacitor, or a capacitor and an inductor. One end of the second sub-branch is electrically connected to the first feeding point, the other end of the second sub-branch is electrically connected to the third reference ground, and the second sub-branch includes an inductor, or an inductor and a capacitor. The other end of the third sub-branch is electrically connected to the third reference ground, and the third sub-branch includes a capacitor, or a capacitor and an inductor. One end of the fourth sub-branch is electrically connected to the first feeding point, the other end of the fourth sub-branch is electrically connected to one end of the third sub-branch, and the fourth sub-branch includes a capacitor, or a capacitor and an inductor. One end of the fifth sub-branch is electrically connected to said the other end of the fourth sub-branch, the other end of the fifth sub-branch is electrically connected to the first feeding module, and the fifth sub-branch includes an inductor.

(59) In an implementation, the first matching module further includes at least one adjustable branch electrically connected to the third reference ground, the at least one adjustable branch each includes at least one of a switching circuit or an adjustable capacitor, and the at least one adjustable branch is configured to tune the resonant frequency of the first resonant mode and the resonant frequency of the second resonant mode.

(60) In an implementation, a length between the first feeding point and the first ground end is $\frac{1}{3}$ times to 1 time a length of the first radiator.

(61) In an implementation, the antenna assembly is configured to support a fifth resonant mode and a third resonant mode, the fifth resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by excitation current of the second feeding module resonating from the second feeding point to the second coupling end, and the third resonant mode is a $\frac{3}{4}$ wavelength mode caused by the excitation current of the second feeding module resonating on the first radiator.

(62) In an implementation, a resonant frequency of the fifth resonant mode is less than a resonant frequency of the third resonant mode, the fifth resonant mode supports at least one of a Wi-Fi 5 GHz (Wi-Fi 5G) frequency band or a Wi-Fi 6 GHz extended (Wi-Fi 6E) frequency band, and the third resonant mode supports at least one of a Wi-Fi 5G frequency band or a Wi-Fi 6E frequency band.

(63) In an implementation, the second matching module includes at least one third branch connected in parallel to the ground and/or at least one fourth branch connected in series between the first feeding point and the first feeding module, and each of the at least one third branch and the at least one fourth branch includes at least one of a capacitor or an inductor. The second matching module is configured to allow a frequency band supported by the fifth resonant mode and a frequency band supported by the third resonant mode to pass, to block a frequency band less than both the frequency band supported by the fifth resonant mode and the frequency band supported by the third resonant mode, and to tune a resonant frequency of the fifth resonant mode and/or a resonant frequency of the third resonant mode.

(64) In an implementation, the at least one third branch includes a sixth sub-branch and a seventh sub-branch, and the at least one fourth branch includes an eighth sub-branch. One end of the sixth sub-branch is electrically connected to the second feeding point, the other end of the sixth sub-branch is electrically connected to one end of the seventh sub-branch, and the sixth sub-branch includes an inductor and a capacitor connected in parallel. The other end of the seventh sub-branch

is electrically connected to the second feeding module, and the seventh sub-branch includes a capacitor, or an inductor, or an inductor and a capacitor connected in parallel. One end of the eighth sub-branch is electrically connected to the second feeding module, the other end of the eighth sub-branch is electrically connected to a fourth reference ground, and the eighth sub-branch includes at least one of a capacitor or an inductor.

(65) In an implementation, the antenna assembly further includes a third radiator, a third feeding module, and a third matching module. The third radiator has a third ground end, a first free end, and a third feeding point between the third ground end and the first free end. The third ground end and the first ground end are spaced apart from each other or are connected via a conductor. The third ground end is electrically connected to a fifth reference ground, the third feeding module is electrically connected to the third feeding point, and the third matching module is electrically connected between the third feeding point and the third feeding module.

(66) In an implementation, the antenna assembly further includes a fourth matching module, where one end of the fourth matching module is electrically connected to the first ground end, and the other end of the fourth matching module is electrically connected to the first reference ground; and/or the antenna assembly further includes a fifth matching module, where one end of the fifth matching module is electrically connected to the third ground end, and the other end of the fifth matching module is electrically connected to the fifth reference ground.

(67) In an implementation, the antenna assembly further includes a fourth radiator, a fourth feeding module, and a sixth matching module. The fourth radiator has a fourth ground end, a second free end, and a fourth feeding point between the fourth ground end and the second free end. A second coupling gap is defined between the fourth ground end and the first free end, and the fourth ground end is electrically connected to a sixth reference ground. The fourth feeding module is electrically connected to the fourth feeding point, and the sixth matching module is electrically connected between the fourth feeding point and the fourth feeding module.

(68) In an implementation, the antenna assembly is further configured to support a sixth resonant mode, a seventh resonant mode, and an eighth resonant mode. The sixth resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by excitation current of the third feeding module resonating on the third radiator. The seventh resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module resonating from the third feeding point to the first free end. The eighth resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module resonating from the third feeding point to the first free end and is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module resonates from the second free end to the fourth ground end.

(69) The antenna assembly is further configured to support a ninth resonant mode, a tenth resonant mode, an eleventh resonant mode, and a twelfth resonant mode. The ninth resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by excitation current of the fourth feeding module resonating from the third feeding point to the first free end and is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the fourth feeding module resonating from the first free end to the fourth ground end. The tenth resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the fourth feeding module resonating from the first free end to the fourth ground end. The eleventh resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the fourth feeding module resonating from the fourth feeding point to the second free end. The twelfth resonant mode is a $\frac{3}{4}$ wavelength mode caused by the excitation current of the fourth feeding module resonating from the third ground end to the first free end.

(70) In an implementation, the third radiator supports, under excitation of the third feeding module, a GPS frequency band, a Wi-Fi 2.4G frequency band, an LTE-4G MHB frequency band, an NR-5G MHB frequency band, an LTE-4G UHB frequency band, and an NR-5G UHB frequency band. The fourth radiator supports, under excitation of the fourth feeding module, an N77 (3.3 GHz~4.2 GHz) frequency band, an N78 (3.3 GHz~3.8 GHz) frequency band, a Wi-Fi 5G frequency band, and a

Wi-Fi 6E frequency band.

(71) In an implementation, the first radiator is connected to the third radiator to form a sharing radiator. The antenna assembly further includes a direct-current (DC) blocking component, a filtering component, and a detecting component. The DC blocking component is electrically connected between the first feeding point and the first matching module, between the first ground end and the first reference ground, between the third ground end and the fifth reference ground, and between the third feeding point and the third matching module; one end of the filtering component is electrically connected to one end of the DC blocking component close to the sharing radiator or is electrically connected to the sharing radiator; the DC blocking component is configured to block direct current generated by the first matching module, the first reference ground, the fifth reference ground, and the third matching module; and the filtering component is configured to block a radio frequency (RF) signal received/transmitted by the sharing radiator and to block a sensing signal generated by the sharing radiator in response to approach of a subject to-be-detected. And/or the DC blocking component is electrically connected between the second ground end and the second reference ground, and between the second matching module and the second feeding point; one end of the filtering component is electrically connected to one end of the DC blocking component close to the second radiator or is electrically connected to the second radiator; the DC blocking component is configured to block direct current generated by the second reference ground and the second matching module; and the filtering component is configured to block an RF signal received/transmitted by the second radiator and to block a sensing signal generated by the second radiator in response to the approach of the subject to-be-detected. And/or the DC blocking component is electrically connected between the fourth ground end and the sixth reference ground, and between the sixth matching module and the fourth feeding point; one end of the filtering component is electrically connected to one end of the DC blocking component close to the third radiator or is electrically connected to the third radiator; the DC blocking component is configured to block direct current generated by the sixth reference ground and the sixth matching module; and the filtering component is configured to block an RF signal received/transmitted by the third radiator and to block a sensing signal generated by the third radiator in response to the approach of the subject to-be-detected. The detecting component is electrically connected to the other end of the filtering component, and the detecting component is configured to detect magnitude of the sensing signal.

(72) In an implementation, a segment between the first ground end and the third ground end is a connecting segment. The antenna assembly further includes at least one seventh matching module, one end of the at least one seventh matching module is electrically connected to the connecting segment, and the other end of the seventh matching module is connected to the ground.

(73) In a second aspect, an electronic device is provided in the present disclosure. The electronic device includes a housing and at least one antenna assembly of the first aspect or any one of the first aspect, and radiators of the at least one antenna assembly are integrated into the housing, or are disposed on a surface of the housing, or are disposed in space enclosed by the housing.

(74) In an implementation, the housing has a top frame and a bottom frame disposed opposite to the top frame and a first side frame and a second side frame connected between the top frame and the bottom frame. The antenna assembly is disposed on at least one of the top frame, the first side frame, the second side frame, or the bottom frame.

(75) In an implementation, a first radiator of the antenna assembly and a second radiator of the antenna assembly are disposed on the first side frame, one part of a third radiator of the antenna assembly is disposed on the first side frame, and the other part of the third radiator of the antenna assembly is disposed on the top frame. A fourth radiator of the antenna assembly is disposed on the top frame. A first coupling gap between the first radiator and the second radiator is defined on the first side frame, and a second coupling gap is defined on the top frame.

(76) The electronic device further includes at least one key portion, the at least one key portion is

disposed between a second ground end and a second feeding point, and/or the at least one key portion is disposed between a first ground end and a third ground end of the third radiator.

(77) The following will clearly and completely illustrate technical solutions of embodiments of the disclosure with reference to the accompanying drawings of embodiments of the disclosure. Apparently, embodiments described herein are merely some embodiments, rather than all embodiments, of the disclosure. In addition, reference herein to “implementation” or “embodiment” means that a particular feature, structure, or characteristic described in connection with the implementation or embodiment may be included in at least one embodiment of the present disclosure. An appearance of this term in various places in the specification does not necessarily refer to the same embodiment, nor is a separate or alternative embodiment mutually exclusive of other embodiments. It is understood explicitly and implicitly by those skilled in the art that embodiments described herein can be combined with other embodiments.

(78) Reference is made to FIG. 1, which is a schematic structural diagram of an electronic device provided in embodiments of the present disclosure. The electronic device **1000** includes an antenna assembly **100**. The antenna assembly **100** is configured to receive and transmit electromagnetic wave signals to implement a communication function of the electronic device **1000**. A position of the antenna assembly **100** in the electronic device **1000** is not specifically limited in the present disclosure. The electronic device **1000** further includes a display screen **300** and a housing **200**, and the display screen **300** covers and is connected to the housing **200**. The antenna assembly **100** may be disposed inside the housing **200** of the electronic device **1000**, or partially integrated with the housing **200**, or partially disposed outside the housing **200**. As illustrated in FIG. 1, a radiator of the antenna assembly **100** is integrated with the housing **200**. Certainly, the antenna assembly **100** may also be disposed on a retractable assembly of the electronic device **1000**. In other words, at least part of the antenna assembly **100** can also extend out of the electronic device **1000** along with the retractable assembly of the electronic device **1000**, and retract into the electronic device **1000** along with the retractable assembly. Alternatively, an overall length of the antenna assembly **100** may extend with extension of the retractable assembly of the electronic device **1000**.

(79) The electronic device **1000** includes, but is not limited to, a device capable of receiving/transmitting the electromagnetic wave signal, such as a telephone, a television, a tablet computer, a mobile phone, a camera, a personal computer (PC), a notebook computer, an in-vehicle equipment, an earphone, a watch, a wearable device, a base station, an in-vehicle radar, and a customer premise equipment (CPE). In the present disclosure, for illustrative purpose, the electronic device **1000** is the mobile phone, and for other devices, reference may be made to detailed descriptions in this disclosure.

(80) For ease of description, with reference to a view angle of the electronic device **1000** in FIG. 1, a width direction of the electronic device **1000** is defined as an X-axis direction, a length direction of the electronic device **1000** is defined as a Y-axis direction, and a thickness direction of the electronic device **1000** is defined as a Z-axis direction. The X-axis, the Y-axis and the Z-axis are perpendicular to each other, and a direction indicated by an arrow is a positive direction.

(81) Reference is made to FIG. 2, where the housing **200** includes a frame **210** and a back cover **220**. A middle plate **410** is formed within the frame **210** by injection molding, and the middle plate **410** defines multiple installation grooves for installing various electronic devices. The middle plate **410** and the frame **210** cooperatively serve as a middle frame **420** of the electronic device **1000**, and accommodating space is defined at both sides of the middle frame **420** after the display screen **300**, the middle frame **420**, and the back cover **220** are closed. One side (for example, a back side) of the frame **210** is connected to a periphery of the back cover **220**, and the other side (for example, the front side) of the frame **210** is connected to a periphery of the display screen **300**. The electronic device **1000** further includes devices capable of implementing basic functions of the mobile phone, such as a battery, a camera, a microphone, a receiver, a loudspeaker, a face-recognition module, a fingerprint-recognition module, or the like, all of which are disposed inside

the accommodating space, and are not further described in this embodiment.

(82) The antenna assembly **100** provided in the present disclosure will be specifically described below with reference to accompanying drawings. Certainly, the antenna assembly **100** provided in the present disclosure includes, but is not limited to, the following embodiments.

(83) As illustrated in FIG. 3, the antenna assembly **100** includes at least a first radiator **11**, a second radiator **21**, a first matching module **M1**, a first feeding module **13**, a second matching module **M2**, and a second feeding module **23**. To facilitate functional division of different parts of the antenna assembly **100** for subsequent descriptions, the first radiator **11**, the second radiator **21**, the first matching module **M1**, and the first feeding module **13** are defined as a first antenna unit **10**, and the first radiator **11**, the second radiator **21**, the second matching module **M2**, and the second feeding module **23** are defined as a second antenna unit **20**.

(84) As illustrated in FIG. 3, the first radiator **11** has a first ground end **111**, a first coupling end **112**, and a first feeding point A between the first ground end **111** and the first coupling end **112**. In this embodiment, the first ground end **111** and the first coupling end **112** are opposite ends of the first radiator **11**, and the first radiator **11** is in a linear strip shape. In other embodiments, the first radiator **11** is bent, and the first ground end **111** and the first coupling end **112** may not be opposite to each other along a straight-line direction, but the first ground end **111** and the first coupling end **112** are two ends of the first radiator **11**.

(85) As illustrated in FIG. 3, the second radiator **21** has a second coupling end **211**, a second ground end **212**, and a second feeding point B between the second coupling end **211** and the second ground end **212**. A first coupling gap **140** is defined between the second coupling end **211** and the first coupling end **112**, and the first radiator **11** and the second radiator **21** can be capacitively coupled via the first coupling gap **140**. In this embodiment, the second coupling end **211** and the second free end **122** are two ends of the second radiator **21**. Optionally, the first radiator **11** and the second radiator **21** may be arranged along a straight line or roughly along a straight line (namely, having a small tolerance in the design process). Certainly, in other embodiments, the first radiator **11** and the second radiator **21** may also be staggered in an extending direction to define avoidance space.

(86) As illustrated in FIG. 3, the first coupling end **112** is opposite to and is spaced apart from the second coupling end **211**. The first coupling gap **140** is a gap between the first radiator **11** and the second radiator **21**. For example, a width of the first coupling gap **140** may range from 0.5 mm to 2 mm, but is not limited to this size. The first radiator **11** and the second radiator **21** may be regarded as two parts formed by a radiator interrupted by the first coupling gap **140**.

(87) The first radiator **11** and the second radiator **21** are capacitively coupled via the first coupling gap **140**. The “capacitively coupled” refers to that an electric field is generated between the first radiator **11** and the second radiator **21**, a signal of the first radiator **11** can be transmitted to the second radiator **21** via the electric field, and a signal of the second radiator **21** can be transmitted to the first radiator **11** via the electric field, so that the first radiator **11** and the second radiator **21** can be electrically conducted even when the first radiator **11** and the second radiator **21** are not directly contacted or are not directly connected.

(88) It can be understood that, shapes and structures of the first radiator **11** and the second radiator **21** are not specifically limited in the present disclosure, and shapes of the first radiator **11** and the second radiator **21** include, but are not limited to, a strip shape, a sheet shape, a rod shape, a coating, a film, or the like. When the first radiator **11** and the second radiator **21** are in the strip shape, the present disclosure does not limit extending track of the first radiator **11** and the second radiator **21**, and therefore the first radiator **11** and the second radiator **21** may both extend in a track such as a straight line, a curve, multiple bends, or the like. The extending track of the first radiator **11** and the second radiator **21** may be a line with a uniform width, or a strip shape with varying widths such as a gradual width, a widened area, or the like.

(89) As illustrated in FIG. 3, the first matching module **M1** is electrically connected between the

first feeding point A and the first feeding module **13**. The first feeding module **13** is an RF receiving/transmitting chip configured to transmit RF signals or a feeding part electrically connected to the RF receiving/transmitting chip configured to transmit RF signals. The first matching module **M1** may include at least one of a switching device, a capacitive device, an inductive device, a resistance device, or the like.

(90) As illustrated in FIG. 3, the first ground end **111** is electrically connected to a first reference ground **GND1** by, including but not limited to, direct welding or indirect electrical connection via coaxial line, microstrip wire, conductive dome, conductive adhesive, or the like. The present disclosure does not limit a specific position of the first feeding point A on the first radiator **11**.

(91) As illustrated in FIG. 3, the second antenna unit **20** includes the first radiator **11**, the second radiator **21**, the second matching module **M2**, and the second feeding module **23**.

(92) As illustrated in FIG. 3, the second matching module **M2** is electrically connected between the second feeding point B and the second feeding module **23**. The second feeding module **23** is an RF receiving/transmitting chip configured to transmit RF signals or a feeding part electrically connected to the RF receiving/transmitting chip configured to transmit RF signals. The second matching module **M2** includes at least one of a switch device, a capacitive device, an inductive device, a resistance device, or the like.

(93) As illustrated in FIG. 3, the second ground end **212** is electrically connected to a second reference ground **GND2** by, including but not limited to, direct welding or indirect electrical connection via coaxial line, microstrip wire, conductive dome, conductive adhesive, or the like. The present disclosure does not limit a specific position of the second feeding point B on the second radiator **21**.

(94) The first reference ground **GND1** and the second reference ground **GND2** include, but are not limited to, the following embodiments. Optionally, the antenna assembly **100** itself has a reference ground. Specific forms of the reference ground include, but are not limited to, a metal conductive plate, a metal conductive layer formed inside a flexible printed circuit board (FPCB) or formed in a rigid printed circuit board (PCB), or the like. The first reference ground **GND1** and the second reference ground **GND2** may be an integrally formed reference ground in the antenna assembly **100**, or two independent reference grounds connected with each other in the antenna assembly **100**. When the antenna assembly **100** is disposed inside the electronic device **1000**, a reference ground of the antenna assembly **100** is electrically connected to a reference ground of the electronic device **1000**. Alternatively, the antenna assembly **100** itself does not have a reference ground, and the first ground end **111** of the antenna assembly **100** and the second ground end **212** of the antenna assembly **100** are directly electrically connected to or are indirectly electrically connected via a conductor to the reference ground of the electronic device **1000** or to a reference ground of electronic components inside the electronic device **1000**. In the present disclosure, the antenna assembly **100** is disposed on the electronic device **1000**, and a metal alloy on the middle plate **410** serves as a reference ground **GND**. In other words, the first reference ground **GND1** and the second reference ground **GND2** each are a part of the middle plate **410**, or are electrically connected to the middle plate **410**, and subsequent third reference ground **GND3**, a fourth reference ground **GND4**, a fifth reference ground **GND5**, a sixth reference ground **GND6**, and a seventh reference ground **GND7** are all part of the reference ground **GND**.

(95) In this embodiment, the first radiator **11** and the second radiator **21** support, under excitation of the first feeding module **13** and the second feeding module **23**, multiple resonant modes. Specifically, the first radiator **11** and the second radiator **21** support, under excitation of the first feeding module **13**, multiple resonant modes (for example, a first resonant mode a, a second resonant mode b, and a fourth resonant mode c in FIG. 4). Specifically, the first radiator **11** and the second radiator **21** support, under excitation of the second feeding module **23**, multiple resonant modes (for example, a fifth resonant mode d and a third resonant mode e in FIG. 4).

(96) Because the first radiator **11** is capacitively coupled to the second radiator **21**, current

generated by the first feeding module **13** can resonate on the first radiator **11** and/or the second radiator **21**, and current generated by the second feeding module **23** can also resonate on the first radiator **11** and/or the second radiator **21**. The first radiator **11** and the second radiator **21** can serve not only as radiators of the first antenna unit **10** but also as radiators of the second antenna unit **20**. Compared with that current generated by the first feeding module **13** separately resonating on the first radiator **11** and current generated by the second feeding module **23** separately resonating on the second radiator **21**, the first antenna unit **10** and the second antenna unit **20** in the antenna assembly **100** provided in embodiments of the present disclosure can implement co-aperture radiation, so as to generate more resonant modes. From other perspectives, the antenna assembly **100** provided in embodiments of the present disclosure fully utilizes radiators while supporting required frequency bands (or supporting the required number of resonant modes), so as to further reduce an overlapping length of the first radiator **11** and the second radiator **21**.

(97) At least one resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the first feeding module **13** resonating on the second radiator **21**. The wavelength refers to a wavelength of an electromagnetic wave in the medium where the electromagnetic wave is located. Specifically, current provided by the first feeding module **13** resonates on the first radiator **11** to generate a resonant mode. In addition, because the first radiator **11** is capacitively coupled to the second radiator **21**, current provided by the first feeding module **13** is further transmitted to the second radiator **21** through the first coupling gap **140**, so as to generate a resonant mode on the second radiator **21**, thereby increasing the number of resonant modes supported by the first antenna unit **10**. Further, a length of the second radiator **21** is designed to be about $\frac{1}{8}$ times to $\frac{1}{4}$ times a wavelength of current provided by the first feeding module **13** in the medium to facilitate generating the resonant mode, so as to make current resonate in a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode on the second radiator **21**. The $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode is a resonant mode with relatively high efficiency, and therefore, the number of resonant modes supported by the first antenna unit **10** can be increased and the receiving/transmitting efficiency for supported frequency bands can be improved. In this way, the first antenna unit **10** can generate resonant modes on both the first radiator **11** and the second radiator **21**. Compared with that current of the first antenna unit **10** only resonates on the first radiator **11**, the number of resonant modes supported by the first antenna unit **10** is increased and signal conversion efficiency becomes higher, so that the number of resonant modes supported by the antenna assembly **100** is increased.

(98) The number of resonant modes supported by the antenna assembly **100** is increased, and a frequency-band width supported by the antenna assembly **100** and the number of frequency bands supported by the antenna assembly **100** are increased. Specifically, in a first aspect, when frequency bands supported by multiple resonant modes of the antenna assembly **100** are all continuous, the frequency-band width supported by the antenna assembly **100** is relatively wide, and a relatively wide bandwidth can be formed, where the relatively wide bandwidth is 1G, 1.5G, 2G, or the like, so that an relatively wide bandwidth coverage can be realized, a download bandwidth can be improved, a throughput download speed can be improved, and a user's Internet experience can be enhanced. In a second aspect, when frequency bands supported by multiple resonant modes of the antenna assembly **100** are discontinuous, the number of frequency bands supported by the antenna assembly **100** is increased, and a multi-band coverage is achieved, for example, the antenna assembly **100** can simultaneously support a 4G/5G MHB (for example, 1000 MHz to 3000 MHz) and a 4G/5G UHB (for example, 3000 MHz to 10000 MHz), simultaneously support MHBs in two different frequency bands, simultaneously support the 4G/5G MHB and a Wi-Fi frequency band (for example, Wi-Fi 5G, 5.925 GHz to 7.125 GHz, or the like), or the like. Frequency bands supported by multiple resonant modes being continuous means that two adjacent frequency bands supported by multiple resonant modes are at least partially overlapped. Frequency bands supported by multiple resonant modes being discontinuous means that no overlap exists between two adjacent bands supported by multiple resonant modes.

(99) When the antenna assembly **100** is configured to cover two signals with different communication protocols (for example, a mobile communication 4G/5G signal and a Wi-Fi signal), the first antenna unit **10** can support relatively more resonant modes, cover a relatively wide frequency-band width, and support a relatively large number of frequency bands, so that the first antenna unit **10** supports one communication protocol signal (for example, the mobile communication 4G/5G signal) that requires support of more resonant modes, and the second antenna unit **20** supports the other communication protocol signal (for example, the Wi-Fi signal). In this way, one antenna unit does not need to support two signals with different communication protocols, and thus devices such as a combiner do not need to be arranged, so as to avoid increased losses on an RF link.

(100) In the antenna assembly **100** and the electronic device **1000** provided in the present disclosure, the first radiator **11** and the second radiator **21** are designed to be capacitively coupled to each other, and the first radiator **11** and the second radiator **21** each serve as a co-aperture radiator of the first antenna unit **10** and the second antenna unit **20**. The length of the second radiator **21** is also designed to correspond to a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength of the resonant frequency of the excitation current provided by the first feeding module **13**, so that current from the first feeding module **13** resonates on the first radiator **11** and resonates on a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode on the second radiator **21** through the first coupling gap **140**. In this way, the number of resonant modes supported by the first antenna unit **10** and receiving/transmitting efficiency for supported frequency bands can be increased, so as to increase the number or bandwidth of frequency bands supported by the first antenna unit **10** and to improve data transmission rate and communication quality. In addition, the second antenna unit **20** can support signals with the same protocol standard, so as to avoid RF link losses generated by receiving/transmitting signals with different protocol standards through one radiator, to make the structure of the antenna unit **100** more simple and compact, and to reduce RF link losses.

(101) As illustrated in FIG. 4, the first antenna unit **10** at least supports a first resonant mode a, a second resonant mode b, and a fourth resonant mode c.

(102) Optionally, the first radiator **11** at least supports, under excitation of the first feeding module **13**, the first resonant mode a and the fourth resonant mode c. The second radiator **21** at least supports, under excitation of the first feeding module **13**, the second resonant mode b.

(103) The resonant mode is indicated by the antenna assembly **100** having higher efficiency of transmitting/receiving electromagnetic waves at and near a resonant frequency, the resonant frequency is a resonant frequency of the resonant mode, and the resonant frequency and vicinity of the resonant frequency form a frequency band supported or covered by the resonant mode. Optionally, in a return loss (RL) curve, an absolute value of a return loss value being greater than or equal to 5 decibels (dB) (which is merely an example and cannot be construed as a limitation on a return loss value for high efficiency in the present disclosure) is taken as a reference value with higher efficiency of transmitting/receiving electromagnetic waves. A set of frequencies whose absolute values of the return loss values in one resonant mode are greater than or equal to 5 dB is taken as a frequency band supported by the resonant mode.

(104) As illustrated in FIG. 5, first-resonant-current density of the first resonant mode a is mainly distributed between the first ground end **111** and the first coupling end **112**, and first resonant current flows from the first ground end **111** to the first coupling end **112**, or flows from the first coupling end **112** to the first ground end **111**. It can be understood that, current density of the first resonant mode a is mainly distributed as described above, and a small amount of current density of the first resonant mode a is distributed on the second radiator **21**.

(105) Specifically, references are made to FIG. 4 and FIG. 5, the first resonant mode a is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current (namely, the first resonant current) of the first feeding module **13** resonating on the first radiator **11**. Specifically, a length of the first radiator **11** is designed to be about $\frac{1}{8}$ times to $\frac{1}{4}$ times a wavelength of the excitation current, in the medium,

flowing from the first feeding module **13** to the first radiator **11**, so that the first resonant mode a is excited on the first radiator **11**, and higher radiation efficiency can be achieved in a frequency band supported by the first resonant mode a.

(106) Generally, the length of the first radiator **11** is designed to be about $\frac{1}{4}$ times the wavelength of the excitation current, in the medium, flowing from the first feeding module **13** to the first radiator **11**, so that the higher radiation efficiency is easily excited at a first resonant frequency point **f1** (namely, a resonant frequency of the first resonant mode a). Further, a capacitive matching circuit connected to the ground is disposed on a path of the first resonant current so as to feed the first resonant current to the first radiator **11** through capacitive coupling. A capacitive loading may enable the resonant frequency of the first resonant mode a to be shifted towards lower frequencies, so that the higher-efficiency resonance is not just generated when the length of the first radiator **11** is about a $\frac{1}{4}$ wavelength, but also generated when the length of the first radiator **11** is in a range of a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength. Therefore, while a resonance is generated at the first resonant frequency point **f1**, the length of the first radiator **11** can also be shortened, for example, shortened to $\frac{1}{8}$ times a wavelength corresponding to the first resonant frequency point **f1**, so as to further reduce the length of the first radiator **11** and to reduce an overlapping length of the antenna assembly **100**.

(107) Optionally, as illustrated in FIG. 6, second-resonant-current density of the second resonant mode b is mainly distributed between the first feeding point A and the second ground end **212**. Second resonant current flows from the first feeding point A to the first coupling end **112**, through the first coupling gap **140** to the second coupling end **211**, and from the second coupling end **211** to the second ground end **212**; or the second resonant current flows from the second ground end **212** to the second coupling end **211**, through the first coupling gap **140** to the first coupling end **112**, and from the first coupling end **112** to the first feeding point A. It can be understood that, current density of the second resonant mode b is mainly distributed as described above, and a small amount of current density of the second resonant mode b is distributed on the second radiator **21**.

(108) References are made to FIG. 4 and FIG. 6, the second resonant mode b is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current (namely, the second resonant current) of the first feeding module **13** resonating on the second radiator **21**. Specifically, a length of the second radiator **21** is designed to be about $\frac{1}{8}$ times to $\frac{1}{4}$ times a wavelength of the excitation current, in the medium, flowing from the first feeding module **13** to the second radiator **21**, so as to excite the second resonant mode b on the second radiator **21**.

(109) Generally, the length of the second radiator **21** is designed to be about $\frac{1}{4}$ times the wavelength of the excitation current, in the medium, flowing from the first feeding module **13**, so that the higher radiation efficiency is easily excited at a second resonant frequency point **f2** (namely, a resonant frequency of the second resonant mode b). Further, a capacitive matching circuit connected to the ground is disposed on a path of the second resonant current, so as to feed the second resonant current to the second radiator **21** through capacitive coupling. The capacitive loading may enable the resonant frequency of the second resonant mode b to be shifted towards lower frequencies, so that the higher-efficiency resonance is not just generated when the length of the second radiator **21** is about $\frac{1}{4}$ wavelength, but also generated when the length of the second radiator **21** is in the range of the $\frac{1}{8}$ to $\frac{1}{4}$ wavelength. Therefore, while the resonance is generated at the second resonant frequency point **f2**, the length of the second radiator **21** can also be shortened, for example, shortened to $\frac{1}{8}$ times a wavelength corresponding to the second resonant frequency point **f2**, so as to further reduce the length of the second radiator **21** and to reduce the overlapping length of the antenna assembly **100**.

(110) Optionally, as illustrated in FIG. 7, third-resonant-current density of the fourth resonant mode c is mainly distributed between the first feeding point A and the first coupling point **112**, and third resonant current flows from the first feeding point A to the first coupling point **112**, or flows from the first coupling point **112** to the first feeding point A. It can be understood that, current density of the fourth resonant mode c is mainly distributed as described above, and a small amount of current

density of the fourth resonant mode c is distributed on the second radiator **21** and distributed between the first ground end **111** and the first feeding point A.

(111) References are made to FIG. 4 and FIG. 7, the fourth resonant mode c is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current (namely, the third resonant current) of the first feeding module **13** resonating from the first feeding point A to the first coupling end **112**. Specifically, a length from the first feeding point A to the first coupling end **112** is designed to be about $\frac{1}{8}$ to $\frac{1}{4}$ times the wavelength of the excitation current, in the medium, flowing from the first feeding module **13** to the first radiator **11**, so as to excite the fourth resonant mode c from the first feeding point A to the first coupling end **112**.

(112) Generally, the length from the first feeding point A to the first coupling end **112** is designed to be about $\frac{1}{4}$ times the wavelength of the excitation current, in the medium, flowing from the first feeding module **13**, so that the higher radiation efficiency is easily excited at a third resonant frequency point **f3** (namely, a resonant frequency of the fourth resonant mode c).

(113) Further, the capacitive matching circuit connected to the ground is disposed on a path of the third resonant current, so as to feed the third resonant current to the first radiator **11** through capacitive coupling. The capacitive loading may enable the resonant frequency of the fourth resonant mode c to be shifted towards lower frequencies, so that the higher-efficiency resonance is not just generated when the length from the first feeding point A to the first coupling end **112** is about $\frac{1}{4}$ wavelength, but also generated when the length from the first feeding point A to the first coupling end **112** is in the range of the $\frac{1}{8}$ to $\frac{1}{4}$ wavelength. Therefore, while the resonance is generated at the third resonant frequency point **f3**, the length from the first feeding point A to the first coupling end **112** can also be shortened, for example, shortened to $\frac{1}{8}$ times a wavelength of the third resonant frequency point **f3**, so as to further reduce the length from the first feeding point A to the first coupling end **112** and to reduce the overlapping length of the antenna assembly **100**.

(114) Certainly, in other embodiments, the second resonant mode b is the $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the first feeding module **13** resonating between the first feeding point A and the first coupling end **112**, and the fourth resonant mode c is the $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the first feeding module **13** resonating from the first feeding point A to the first coupling end **112**.

(115) The length of the first radiator **11**, the length from the first feeding point A to the first coupling end **112**, and the length of the second radiator **21** are designed to make the first feeding module **13** excite resonant modes on the first radiator **11**, between the first feeding point A and the first coupling end **112** on the first radiator **11**, and on the second radiator **21**. In this way, the number of resonant modes supported by the first antenna unit **10** is increased, the number of frequency bands supported by the antenna assembly **100** and the frequency-band width supported by the antenna assembly **100** are effectively increased, and data transmission rate of the antenna assembly **100** is increased.

(116) Optionally, the first resonant mode a may support the GPS frequency band, the LTE-4G frequency band, the NR-5G frequency band, the Wi-Fi 2.4G frequency band, the Wi-Fi 5G frequency band, the Wi-Fi 6E frequency band, or a combination of the LTE-4G frequency band and the NR-5G frequency band. The second resonant mode b may support the GPS frequency band, the LTE-4G frequency band, the NR-5G frequency band, the Wi-Fi 2.4G frequency band, the Wi-Fi 5G frequency band, the Wi-Fi 6E frequency band, or the combination of the LTE-4G frequency band and the NR-5G frequency band. The fourth resonant mode c may support the GPS frequency band, the LTE-4G frequency band, the NR-5G frequency band, the Wi-Fi 2.4G frequency band, the Wi-Fi 5G frequency band, the Wi-Fi 6E frequency band, or the combination of the LTE-4G frequency band and the NR-5G frequency band. One resonant mode may support a separate LTE-4G frequency band, a separate NR-5G frequency band, a separate Wi-Fi frequency band, or a combination of the LTE-4G frequency band and the NR-5G frequency band.

(117) In the present disclosure, the magnitude of the resonant frequency in the first resonant mode

a, the magnitude of the resonant frequency in the second resonant mode b, and the magnitude of the resonant frequency in the fourth resonant mode c are not specifically limited. In this embodiment, the resonant frequency of the first resonant mode a, the resonant frequency of the second resonant mode b, and the resonant frequency of the fourth resonant mode c increase sequentially. When the first resonant mode a, the second resonant mode b, and the fourth resonant mode c are all the $\frac{1}{4}$ wavelength mode, since the resonant frequency of the first resonant mode a, the resonant frequency of the second resonant mode b, and the resonant frequency of the fourth resonant mode c increase sequentially, the $\frac{1}{4}$ wavelength corresponding to the resonant frequency of the first resonant mode a, the $\frac{1}{4}$ wavelength corresponding to the resonant frequency of the second resonant mode b, and the $\frac{1}{4}$ wavelength corresponding to the resonant frequency of the fourth resonant mode c decrease sequentially. Compared with disposing a radiating portion of the $\frac{1}{4}$ wavelength corresponding to the resonant frequency of the fourth resonant mode c on the first radiator **11**, a total length of a co-aperture radiator is shorter, by disposing a radiating portion of the $\frac{1}{4}$ wavelength corresponding to the resonant frequency of the second resonant mode b on the first radiator **11**. In this way, the overlapping length of the first radiator **11** and the second radiator **21** of the antenna assembly **100** is shorter, and the space occupied by the antenna assembly **100** is further reduced in extremely compact space.

(118) In this embodiment, the first resonant mode a supports at least one of the GPS frequency band, the LTE-4G MHB frequency band, or the NR-5G MHB frequency band, for example, at least one of a GPS-L1 frequency band, a B3 frequency band, a BI frequency band, an N3 frequency band, or an NI frequency band. For example, the resonant frequency of the first resonant mode a is 1.7698 GHz. The second resonant mode b supports at least one of the Wi-Fi 2.4G frequency band, the LTE-4G MHB frequency band, or the NR-5G MHB frequency band, for example, Wi-Fi 2.4G frequency band, B7 frequency band, B41 frequency band, B38 frequency band, N7 frequency band, N41 frequency band, or the like. For example, the resonant frequency of the second resonant mode b is 2.6185 GHz. The fourth resonant mode c supports at least one of the LTE-4G UHB frequency band, the NR-5G UHB frequency band, or the like, for example, at least one of the N78 frequency band, a B42 frequency band, a B43 frequency band, or the like. For example, the resonant frequency of the fourth resonant mode c is 3.5983 GHz. MHB refers to middle-high frequency band (1000 MHz to 3000 MHz), and UHB refers to ultra-high frequency band (3000 MHz to 10000 MHz).

(119) Optionally, as illustrated in FIG. 4, the second radiator **21** at least supports a fifth resonant mode d and a third resonant mode e.

(120) Optionally, the second radiator **21** between the second coupling end **211** and the second feeding point B at least supports the fifth resonant mode d under excitation of the second feeding module **23**. The first radiator **11** at least supports the third resonant mode e under excitation of the second feeding module **23**.

(121) Optionally, as illustrated in FIG. 4 and FIG. 8, fourth-resonant-current density of the fifth resonant mode d is mainly distributed between the second feeding point B and the second coupling point **211**, and fourth resonant current flows from the second feeding point B to the second coupling point **211**, or flows from the second coupling point **211** to the second feeding point B. It can be understood that, current density of the fifth resonant mode d is mainly distributed as described above, and a small amount of current density of the fifth resonant mode d is distributed on the first radiator **11** and distributed between the second ground end **212** and the second feeding point B.

(122) Specifically, the fifth resonant mode d is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current (namely, the fourth resonant current) of the second feeding module **23** resonating from the second feeding point B to the second coupling end **211**. Specifically, a length from the second feeding point B to the second coupling end **211** is designed to be about $\frac{1}{8}$ times to $\frac{1}{4}$ times a wavelength of the excitation current, in the medium, flowing from the second feeding module **23** to

the second radiator **21**, so as to excite the fifth resonant mode d from the second feeding point B to the second coupling end **211**.

(123) Generally, the length from the second feeding point B to the second coupling end **211** is designed to be about $\frac{1}{4}$ times the wavelength of the excitation current, in the medium, flowing from the second feeding module **23**, so that the higher radiation efficiency is easily excited at a fourth resonant frequency point **f4** (namely, a resonant frequency of the fifth resonant mode d). Further, the capacitive matching circuit connected to the ground is disposed on a path of the fourth resonant current so as to feed the fourth resonant current to the second radiator **21** through capacitive coupling. The capacitive loading may enable the resonant frequency of the fifth resonant mode d to be shifted towards lower frequencies, so that the higher-efficiency resonance is not just generated when the length from the second feeding point B to the second coupling end **211** is about the $\frac{1}{4}$ wavelength, but also generated when the length from the second feeding point B to the second coupling end **211** is in the range of the $\frac{1}{8}$ to $\frac{1}{4}$ wavelength. Therefore, while a resonance is generated at the fourth resonant frequency point **f4**, the length from the second feeding point B to the second coupling end **211** can also be shortened, for example, shortened to $\frac{1}{8}$ times a wavelength corresponding to the fourth resonant frequency point **f4**, so as to further reduce the length from the second feeding point B to the second coupling end **211** and to reduce the overlapping length of the antenna assembly **100**.

(124) Optionally, as illustrated in FIG. **9**, fifth-resonant-current density in the third resonant mode e is mainly distributed between the first ground end **111** and the first coupling end **112**, and fifth resonant current flows from the first ground end **111** to the first coupling end **112**, or flows from the first coupling end **112** to the first ground end **111**. It can be understood that, current density of the third resonant mode e is mainly distributed as described above, and a small amount of current density of the third resonant mode e is distributed between the second coupling end **211** and the second ground end **212**.

(125) The third resonant mode e is a $\frac{3}{4}$ wavelength mode caused by the excitation current of the second feeding module **23** resonating on the first radiator **11**. Specifically, the length of the first radiator **11** is designed to be about $\frac{3}{4}$ times a wavelength of the excitation current, in the medium, flowing from the second feeding module **23** to the first radiator **11**, so as to excite the third resonant mode e on the first radiator **11**.

(126) It should be noted that, the first resonant mode a to the third resonant mode e are described above from the perspective of the wavelength mode, which can be better understood and explained and illustrates main characteristics of each resonant mode so that each resonant mode is easily distinguished. However, when each resonant mode operates, the first antenna unit **10** and the second antenna unit **20** are not independent, but are coupled to each other, and current also flows to each other through the coupling.

(127) It can be understood that, in the second antenna unit **20**, the first radiator **11** is similar to a parasitic radiator of the second radiator **21**. The third resonant mode e is a resonant mode generated when the second feeding module **23** excites the parasitic radiator (of the second radiator **21**).

(128) The length of the first radiator **11** and the length from the second feeding point B to the second coupling end **211** are designed to make the second feeding module **23** excite resonant modes on the first radiator **11** and between the second feeding point B and the second coupling end **211** on the second radiator **21**. In this way, the number of resonant modes supported by the first antenna unit **10** is increased, the number of frequency bands supported by the second antenna unit **20** and the frequency-band width supported by the second antenna unit **20** are effectively increased, and the data transmission rate of the antenna assembly **100** is increased.

(129) Optionally, the fifth resonant mode d may support the GPS frequency band, the LTE-4G frequency band, the NR-5G frequency band, the Wi-Fi 2.4G frequency band, the Wi-Fi 5G frequency band, the Wi-Fi 6E frequency band, or a combination of the LTE-4G frequency band and the NR-5G frequency band. The third resonant mode may support the GPS frequency band, the

LTE-4G frequency band, the NR-5G frequency band, the Wi-Fi 2.4G frequency band, the Wi-Fi 5G frequency band, the Wi-Fi 6E frequency band, or the combination of the LTE-4G frequency band and the NR-5G frequency band. One resonant mode may support the separate LTE-4G frequency band, the separate NR-5G frequency band, the separate Wi-Fi frequency band, or the combination of the LTE-4G frequency band and the NR-5G frequency band.

(130) In the present disclosure, the magnitude of the resonant frequency of the fifth resonant mode d and the magnitude of the resonant frequency of the third resonant mode e are not limited. In this embodiment, the resonant frequency of the fifth resonant mode d is less than the resonant frequency of the third resonant mode e.

(131) Further, the resonant frequency of the fifth resonant mode d is greater than the resonant frequency of the fourth resonant mode c, and $\frac{1}{4}$ wavelength of the resonant frequency of the fifth resonant mode d is less than $\frac{1}{4}$ wavelength of the resonant frequency of the fourth resonant mode c. In the second radiator **21**, a position of the second feeding point B is properly designed, so that a length between the second feeding point B and the second coupling end **211** corresponds to $\frac{1}{4}$ wavelength of the resonant frequency of the fifth resonant mode d, so as to excite the resonant mode between the second feeding point B and the second coupling end **211** on the second radiator **21**.

(132) When the resonant frequency of the third resonant mode e is 3 times to 6 times the resonant frequency of the first resonant mode a (for example, the first resonant mode a supports the GPS-L1 frequency band, the B3 frequency band, the BI frequency band, the N3 frequency band, or the NI frequency band; and the third resonant mode e supports the Wi-Fi 5G frequency band or the Wi-Fi 6E frequency band), the second feeding module **23** may excite a $\frac{3}{4}$ wavelength mode on the first radiator **11**. In this way, the first radiator **11** is also reused while the number of resonant modes is increasing.

(133) All of the above describe a reasonable design of a size of the first radiator **11** and the second radiator **21** and location planning of feeding points, so as to generate, in a limited length of the first radiator **11** and the second radiator **21**, as many resonant modes with relatively high efficiency and frequency bands needed to be supported as possible.

(134) In this embodiment, the fifth resonant mode d supports at least one of the Wi-Fi 5G frequency band or the Wi-Fi 6E frequency band, and the third resonant mode supports at least one of the Wi-Fi 5G frequency band or the Wi-Fi 6E frequency band. For example, the resonant frequency of the fifth resonant mode d is 5.4751 GHz and the resonant frequency of the third resonant mode e is 6.0064 GHz.

(135) In this embodiment, the first antenna unit **10** may support the GPS-L1 frequency band, the Wi-Fi 2.4G frequency band, the LTE-4G MHB frequency band (for example, the B3 frequency band, the BI frequency band, the B7 frequency band, the B41 frequency band, or B38 frequency band), the NR-5G MHB frequency band (for example, the N3 frequency band, the NI frequency band, the N7 frequency band, or the N41 frequency band), the LTE-4G UHB frequency band (for example, the B42 frequency band or the B43 frequency band), and the NR-5G UHB frequency band (for example, the N78 frequency band). The second antenna unit **20** may support the Wi-Fi 5G frequency band, the Wi-Fi 6E frequency band, or the like, such that the second antenna unit **20** only needs to support a communication signal of one communication protocol. In this way, devices such as a combiner, configured to combine two communication signals of different communication protocols, do not need to be disposed in an RF link of the second antenna unit **20**, so that the number of devices in the RF link is reduced, and RF link losses of the antenna assembly **100** are reduced.

(136) In this embodiment, the first matching module M1 is disposed between the first feeding point A and the first feeding module **13**, and the first matching module M1 is configured to select frequency bands from signals fed by the first feeding module **13**. For example, the first matching module M1 selects a frequency band of 1 GHz to 7 GHz from RF signals transmitted from the first

feeding module **13** and transmits the frequency band of 1 GHz to 7 GHz to the first feeding point A, or the first matching module **M1** selects a frequency band of 1 GHz to 4.5 GHz from RF signals transmitted from the first feeding module **13** and transmits the frequency band of 1 GHz to 4.5 GHz to the first feeding point A. In addition, the first matching module **M1** can also tune resonant modes supported by the second antenna unit **20** to make the antenna assembly **100** resonate at frequency bands needed to be supported, and can isolate resonant signals of the second antenna unit **20** to increase isolation between the first antenna unit **10** and the second antenna unit **20**.

(137) The first matching module **M1** will be specifically described below with reference to accompanying drawings.

(138) As illustrated in FIG. **10**, the first matching module **M1** includes at least one first branch **14** connected in parallel to the ground and/or at least one second branch **15** connected in series between the first feeding point A and the first feeding module **13**, and each of the at least one first branch **14** and the at least one second branch **15** includes at least one of a capacitor or an inductor.

(139) The first branch **14** includes, but is not limited to, at least one of a single-capacitor branch, a single-inductor branch, or frequency-selective filter circuits listed in the following embodiments.

(140) As illustrated in FIG. **11**, the first branch **14** includes a band-pass circuit formed by an inductor **L0** and a capacitor **C0** connected in series.

(141) As illustrated in FIG. **12**, the first branch **14** includes a band-stop circuit formed by an inductor **L0** and a capacitor **C0** connected in parallel.

(142) As illustrated in FIG. **13**, the first branch **14** includes a band-pass or band-stop circuit formed by an inductor **L0**, a first capacitor **C1**, and a second capacitor **C2**. The inductor **L0** is connected in parallel with the first capacitor **C1**, and the second capacitor **C2** is electrically connected to a node at which the inductor **L0** is electrically connected to the first capacitor **C1**.

(143) As illustrated in FIG. **14**, the first branch **14** includes a band-pass or band-stop circuit formed by a capacitor **C0**, a first inductor **L1**, and a second inductor **L2**. The capacitor **C0** is connected in parallel with the first inductor **L1**, and the second inductor **L2** is electrically connected to a node where the capacitor **C0** is electrically connected to the first inductor **L1**.

(144) As illustrated in FIG. **15**, the first branch **14** includes a band-pass or band-stop circuit formed by an inductor **L0**, a first capacitor **C1**, and a second capacitor **C2**. The inductor **L0** is connected in series with the first capacitor **C1**, one end of the second capacitor **C2** is electrically connected to one end of the inductor **L0** that is not connected to the first capacitor **C1**, and the other end of the second capacitor **C2** is electrically connected to one end of the first capacitor **C1** that is not connected to the inductor **L0**.

(145) As illustrated in FIG. **16**, the first branch **14** includes a band-pass or band-stop circuit formed by a capacitor **C0**, a first inductor **L1**, and a second inductor **L2**. The capacitor **C0** is connected in series with the first inductor **L1**, one end of the second inductor **L2** is electrically connected to one end of the capacitor **C0** that is not connected to the first inductor **L1**, and the other end of the second inductor **L2** is electrically connected to one end of the first inductor **L1** that is not connected to the capacitor **C0**.

(146) As illustrated in FIG. **17**, the first branch **14** includes a first capacitor **C1**, a second capacitor **C2**, a first inductor **L1**, and a second inductor **L2**. The first capacitor **C1** is connected in parallel with the first inductor **L1**, the second capacitor **C2** is connected in parallel with the second inductor **L2**, and one end of a whole formed by the second capacitor **C2** and the second inductor **L2** connected in parallel is electrically connected to one end of a whole formed by the first capacitor **C1** and the first inductor **L1** connected in parallel.

(147) As illustrated in FIG. **18**, the first branch **14** includes a first capacitor **C1**, a second capacitor **C2**, a first inductor **L1**, and a second inductor **L2**. The first capacitor **C1** and the first inductor **L1** are connected in series to form a first unit **101**, the second capacitor **C2** and the second inductor **L2** are connected in series to form a second unit **102**, and the first unit **101** and the second unit **102** are connected in parallel.

(148) The second branch **15** includes, but is not limited to, at least one of a single-capacitor ground branch, a single-inductor ground branch, or ground branches listed in any embodiment of FIG. **11** to FIG. **18**.

(149) The first matching module **M1** is configured to tune at least one of the resonant frequency of the first resonant mode a, the resonant frequency of the second resonant mode b, or the resonant frequency of the fourth resonant mode c. For example, when the antenna assembly **100** is configured to support frequency bands of LTE/NR-MHB+Wi-Fi-2.4 GHz+N78, a structure of the first matching module **M1** is set, and values of each device are designed, so that the first resonant mode a and the second resonant mode b can support the LTE/NR-MHB frequency band, the second resonant mode b can support the Wi-Fi-2.4 GHz frequency band, and the fourth resonant mode c can support the N78 frequency band.

(150) For example, as illustrated in FIG. **10**, the at least one first branch **14** includes a first sub-branch **141**, a second sub-branch **142**, and a third sub-branch **143**. One end of the first sub-branch **141** is electrically connected to the first feeding point A, and the other end of the first sub-branch **141** is electrically connected to the third reference ground GND3. The first sub-branch **141** includes a single-capacitor sub-branch or a sub-branch with a capacitor and an inductor connected in series. In this embodiment, the first sub-branch **141** is a single-capacitor sub-branch, and the capacitor is marked as C11.

(151) In this embodiment, as illustrated in FIG. **10**, one end of the second sub-branch **142** is electrically connected to the first feeding point A, and the other end of the second sub-branch **142** is electrically connected to the third reference ground GND3. The second sub-branch **142** includes an inductor, or an inductor and a capacitor. In this embodiment, the second sub-branch **142** is a single-inductor sub-branch, and the inductor is marked as L11. In other embodiments, the second sub-branch **142** may also be a sub-branch with an inductor and a capacitor connected in series.

(152) As illustrated in FIG. **10**, the other end of the third sub-branch **143** is electrically connected to the third reference ground GND3. The third sub-branch **143** includes a capacitor, or a capacitor and an inductor. In this embodiment, the third sub-branch **143** is a single-capacitor sub-branch, and the capacitor is marked as C13. In other embodiments, the third sub-branch **143** may also be a sub-branch with an inductor and a capacitor connected in series.

(153) As illustrated in FIG. **10**, the at least one second branch **15** includes a fourth sub-branch **151** and a fifth sub-branch **152**.

(154) As illustrated in FIG. **10**, one end of the fourth sub-branch **151** is electrically connected to the first feeding point A, and the other end of the fourth sub-branch **151** is electrically connected to one end of the third sub-branch **143** away from the third reference ground GND3. The fourth sub-branch **151** includes a capacitor, or a capacitor and an inductor. In this embodiment, the fourth sub-branch **151** is a capacitor, which is marked as C12. In other embodiments, the fourth sub-branch **151** may also be a sub-branch with an inductor and a capacitor connected in series.

(155) As illustrated in FIG. **10**, one end of the fifth sub-branch **152** is electrically connected to said the other end of the fourth sub-branch **151**, and the other end of the fifth sub-branch **152** is electrically connected to the first feeding module **13**. The fifth sub-branch **152** includes an inductor. In this embodiment, the fifth sub-branch **152** is an inductor, which is marked as L12.

(156) The above is merely an example of an embodiment of the first matching module **M1**, and in other embodiments, a sub-branch may be additionally disposed between the first sub-branch **141** and the first feeding point A, or a sub-branch may be additionally disposed between the fourth sub-branch **151** and the first feeding point A.

(157) As illustrated in FIG. **3**, FIG. **4**, and FIG. **10**, the first sub-branch **141**, the third sub-branch **143**, and the fifth sub-branch **152** are configured to tune the second resonant mode b and the fourth resonant mode c. The second sub-branch **142** and the fourth sub-branch **151** are configured to tune the first resonant mode a. Specifically, the first sub-branch **141** (C11) is configured to tune a resonant point of the second resonant mode b and a resonant point of the fourth resonant mode c, so

as to enable the fourth resonant mode c to cover the N78 frequency band and enable the second resonant mode b to cover the N41 frequency band. In addition, the first sub-branch **141** (**C11**) can also isolate the Wi-Fi-5G frequency band and Wi-Fi-6E frequency band in the second antenna unit **20**, so as to prevent the first antenna unit **10** from being interrupted by the Wi-Fi-5G frequency band and Wi-Fi-6E frequency band, and to improve the isolation between the first antenna unit **10** and the second antenna unit **20**. The third sub-branch **143** (**C13**) can tune the second resonant mode b and the fourth resonant mode c, and the fifth sub-branch **152** (**L12**) can tune the second resonant mode b and the fourth resonant mode c. For example, **C11**=1 picofarad (pF), **L11**=4.3 nanohenry (nH), **C12**=1 pF, **C13**=0.5 pF, **L12**=3.3 nH, so that the first resonant mode can efficiently support the LTE/NR-MHB frequency band, the second resonant mode b can efficiently support the Wi-Fi 2.4G frequency band and the N41 frequency band, and the fourth resonant mode c can efficiently support the N78 frequency band.

(158) As illustrated in FIG. **19**, the first matching module **M1** further includes at least one adjustable branch **T** electrically connected to the third reference ground **GND3**. A position of the at least one adjustable branch **T** is not specifically limited in the present disclosure, and the at least one adjustable branch **T** may be the first sub-branch **141**, or the second sub-branch **142**, or the third sub-branch **143**, or newly additionally disposed between the first feeding point **A** and the first sub-branch **141**, or newly additionally disposed between the first sub-branch **141** and the second sub-branch **142**, or newly additionally disposed between the third sub-branch **143** and the fourth sub-branch **144**, or the like. The at least one adjustable branch **T** is configured to tune the resonance frequency of the first resonant mode a and the resonance frequency of the fourth resonant mode c. Specifically, the at least one adjustable branch **T** is configured to adjust a position of a resonant frequency point of the first resonant mode a and a position of a resonant frequency point of the fourth resonant mode c, so that the first resonant mode a and the fourth resonant mode c can support a larger range of required frequency bands. For example, the first matching module **M1** provided with the adjustable branch **T** shifts the resonant frequency of the first resonant mode a from 1.77 GHz to 1.92 GHz, and shifts the resonant frequency of the fourth resonant mode c from 3.6 GHz to 3.7 GHz, so that a range of frequency bands that the antenna assembly **100** can support can be covered.

(159) The present disclosure does not limit a specific structure of the adjustable branch **T**. Optionally, the adjustable branch **T** includes at least one of a switching circuit or an adjustable capacitor. The switching circuit includes, but is not limited to, multiple sub-branches each having a switch and a capacitor, multiple sub-branches each having a switch and a resistor, multiple sub-branches each having a switch and an inductor, and multiple sub-branches each having a switch, an inductor, and a capacitor.

(160) For example, as illustrated in FIG. **20**, the adjustable branch **T** is a part of the second branch **15**. Specifically, the adjustable branch **T** includes a switch and multiple inductors, where the multiple inductors are connected in parallel and have different inductance.

(161) As illustrated in FIG. **21**, the inductance of **L111** is 4.3 nH, and the inductance of **L112** is 3 nH. The switch may be, but is not limited to, a single-pole multi-throw (SPMT) switch or a multi-pole multi-throw (MPMT) switch. As illustrated in FIG. **21**, **S21** is a curve of S-parameters of the first antenna unit **10** when the switch conducts the **L111** to the first matching module **M1**; **S21'** is a S-parameters of the first antenna unit **10** when the switch conducts the **L112** to the first matching module **M1**; **S22** is a curve of S-parameters of the second antenna unit **20** when the switch conducts the **L111** to the first matching module **M1**; **S22'** is a curve of S-parameters of the second antenna unit **20** when the switch conducts the **L112** to the first matching module **M1**; **S23** is a curve of isolation between the first antenna unit **10** and the second antenna unit **20** when the switch conducts the **L111** to the first matching module **M1**; and **S23'** is a curve of isolation between the first antenna unit **10** and the second antenna unit **20** when the switch conducts the **L112** to the first matching module **M1**.

(162) When the switch conducts the L**111** to the first matching module M**1**, the resonant frequency of the first resonant mode a is 1.77 GHz, and the resonant frequency of the second resonant mode b is 3.6 GHz. When the switch conducts the L**112** to the first matching module M**1**, the resonant frequency of the first resonant mode a is 1.92 GHz, and the resonant frequency of the second resonant mode b is 3.7 GHz.

(163) When the first matching module M**1** is provided with the adjustable branch T, in this embodiment, a position of the first feeding point A is further designed to make the adjustable branch T not affect or have less influence on the second resonant mode b, the fifth resonant mode d, and the third resonant mode e while tuning the first resonant mode a and the fourth resonant mode c. In this way, the first matching module M**1** tunes the first resonant mode a and the fourth resonant mode c so as to enlarge a range of LTE-4G/NR5G frequency bands supported by the antenna assembly **100** and to improve the performance of each frequency band, and the antenna assembly **100** can be ensured to support a Wi-Fi 2.4G signal, a Wi-Fi 5G signal, and a Wi-Fi 6E signal.

(164) According to the above analysis on wavelength modes of resonances of the second resonant mode b, the fifth resonant mode d, and the third resonant mode e on the first radiator **11** and the second radiator **21**, it can be learned that, tuning at the first feeding point A has a relatively small influence on resonances of the second resonant mode b and the fifth resonant mode d, and has a relatively large influence on the resonance of the third resonant mode e.

(165) For the third resonant mode e, the current density distribution of the first radiator **11** increases first, then decreases, and then increases from the first coupling end **112** to the first ground end **111** (for example, as illustrated in a dotted curve in FIG. **19**), where a point with strong current of the first radiator **11** is about $\frac{1}{3}$ of a length from the first coupling end **112** to the first ground end **111**. A length between the first feeding point A and the first ground end **111** is $\frac{1}{3}$ times to 1 time the length of the first radiator **11**. Optionally, the first feeding point A is disposed at or near the point with strong current of the first radiator **11**.

(166) The first feeding point A is disposed at or near the point with strong current of the first radiator **11**. The length between the first feeding point A and the first ground end **111** is $\frac{1}{3}$ times to 1 time the length of the first radiator **11**. Further, the length between the first feeding point A and the first coupling end **112** is about $\frac{1}{3}$ times the length of the first radiator **11**, so that the first feeding point A can be disposed at the point with strong current. When the first feeding point A is disposed at the point with strong current, the first matching module M**1** is provided with the adjustable branch T, and the adjustable branch T has a relatively small influence on the third resonant mode e. When the first feeding point A is disposed at or near the position with strong current of the first radiator **11**, and the L**11** in the adjustable branch T is switched to the L**12** in the adjustable branch T, a position of a resonant point of the fifth resonant mode d and a position of a resonant point of the third resonant mode e remain unchanged as illustrated in Figure. In this way, the range of LTE-4G/NR5G frequency bands supported by the antenna assembly **100** is increased, and the antenna assembly **100** can be ensured to support the Wi-Fi signal to make the antenna assembly **100** always support the LTE-4G/NR5G signal and the Wi-Fi signal.

(167) In this embodiment, the second matching module M**2** is disposed between the second feeding point B and the second feeding module **23**, and the second matching module M**2** is configured to select frequency bands from signals fed by the second feeding module **23**. For example, the second matching module M**2** selects a frequency band of 1 GHz to 7 GHz from RF signals transmitted from the second feeding module **23** and transmits the frequency band of 1 GHz to 7 GHz to the second feeding point B, or the second matching module M**2** selects a frequency band of 4.5 GHz to 7 GHz from RF signals transmitted from the second feeding module **23** and transmits the frequency band of 4.5 GHz to 7 GHz to the second feeding point B. In addition, the second matching module M**2** can also tune resonant modes supported by the second antenna unit **20** to make the antenna assembly **100** resonate at frequency bands needed to be supported, and can isolate resonant signals of the first antenna unit **10** to increase the isolation between the first antenna unit **10** and the second

antenna unit **20**.

(168) The second matching module **M2** will be specifically described below with reference to accompanying drawings.

(169) As illustrated in FIG. **22**, the second matching module **M2** includes at least one third branch **16** connected in parallel to the ground and/or at least one fourth branch **17** connected in series between the second feeding point **B** and the second feeding module **23**. Each of the at least one third branch **16** and the at least one fourth branch **17** includes at least one of a capacitor or an inductor. The second matching module **M2** is configured to allow frequency bands supported by the fifth resonant mode **d** and frequency bands supported by the third resonant mode **e** (for example, the Wi-Fi 5G frequency band) to pass, to block a frequency band less than both the frequency band supported by the fifth resonant mode **d** and the frequency band supported by the third resonant mode **e** (for example, a frequency band supported by the first antenna unit **10**) to increase the isolation between the first antenna unit **10** and the second antenna unit **20**, and to tune the resonant frequency of the fifth resonant mode **d** and/or the resonant frequency of the third resonant mode **e** to make both the fifth resonant mode **d** and the third resonant mode **e** support the Wi-Fi 5G frequency band and the Wi-Fi 6E frequency band.

(170) The third branch **16** includes, but is not limited to, at least one of a single-capacitor branch, a single-inductor branch, or frequency-selective filter circuits listed in FIG. **11** to FIG. **18**.

(171) The fourth branch **17** includes, but is not limited to, at least one of a single-capacitor branch, a single-inductor branch, or frequency-selective filter circuits listed in FIG. **11** to FIG. **18**.

(172) For example, as illustrated in FIG. **22**, the at least one third branch **16** includes a sixth sub-branch **161** and a seventh sub-branch **162**. One end of the sixth sub-branch **161** is electrically connected to the second feeding point **B**, and the other end of the sixth sub-branch **161** is electrically connected to one end of the seventh sub-branch **162**. The sixth sub-branch **161** includes an inductor and a capacitor connected in parallel, where the inductor is marked as **L21** and the capacitor is marked as **C21**.

(173) The other end of the seventh sub-branch **162** is electrically connected to the second feeding module **23**. The seventh sub-branch **162** includes a capacitor, or an inductor, or an inductor and a capacitor connected in parallel. In this embodiment, the seventh sub-branch **162** is an inductor, and the inductor is marked as **L22**.

(174) As illustrated in FIG. **22**, the at least one fourth branch **17** includes an eighth sub-branch. One end of the eighth sub-branch is electrically connected to the second feeding module **23**, and the other end of the eighth sub-branch is electrically connected to the fourth reference ground **GND4**. The eighth sub-branch includes at least one of a capacitor or an inductor. In this embodiment, the eighth sub-branch is an inductor, and the inductor is marked as **L23**.

(175) The above is merely an example of an embodiment of the second matching module **M2**, and in other embodiments, a branch may be additionally disposed between the sixth sub-branch **161** and the second feeding point **B**.

(176) The sixth sub-branch **161** (namely, **L21** and **C21**) and the seventh sub-branch **162** (namely, **L22**) are configured to block the Wi-Fi 2.4G frequency band, a frequency band of the second resonant mode **b**, and a frequency band of the fourth resonant mode **c**, and to allow the Wi-Fi 5G frequency band and the Wi-Fi 6E frequency band to pass. For example, **L21**=13 nH, **C21**=0.3 pF, and **L22**=2.7 nH. The sixth sub-branch **161** and the seventh sub-branch **162** have low-impedance or band-pass on the Wi-Fi 5G frequency band so as to allow the Wi-Fi 5G frequency band to pass, and can also effectively filter out a frequency band less than the Wi-Fi 5G frequency band so as to reduce the influence of a frequency band signal supported by the first antenna unit **10** on the second antenna unit **20**. The eighth sub-branch **L23** can tune the Wi-Fi 5G and isolate the frequency band supported by the first resonant mode **a**, so as to reduce the influence of a signal supported by the first antenna unit **10** on the second antenna unit **20**. For example, **L23**=1.5 nH.

(177) From a curve **S2**, **1** of isolation between the first antenna unit **10** and the second antenna unit

20 illustrated in FIG. 4, it can be seen that isolation values corresponding to 1 GHz to 3 GHz and 6 GHz to 7 GHz are below -25 , and isolation values illustrated in the figure is also below -15 , which indicates that the first antenna unit **10** has better isolation with the second antenna unit **20**. The first matching module **M1** and the second matching module **M2** described above are designed to make the first antenna unit **10** have better isolation with the second antenna unit **20**, so that efficiency of receiving/transmitting antenna signals of the first antenna unit **10** can be improved, and efficiency of receiving/transmitting antenna signals of the second antenna unit **20** can be improved. It can be seen from FIG. 4 that, after the first matching module **M1** is provided with the adjustable branch **T**, the first antenna unit **10** still has better isolation with the second antenna unit **20**.

(178) The above is the antenna assembly **100** provided in a first type of embodiment. The antenna assembly **100** includes the first antenna unit **10** and the second antenna unit **20** coupled to the first antenna unit **10**, so as to realize a co-aperture of LTE-4G/NR5G/Wi-Fi signals. The antenna assembly **100** supports multiple resonant modes at the same time, and realizes a relatively wide bandwidth design by a carrier-aggregation technology. Meanwhile, the antenna assembly **100** can support an LTE-4G/NR5G dual-connection technology. In the present disclosure, the length of the first radiator **11**, the position of the first feeding point **A**, and the length of the second radiator **21** are reasonably designed, so as to make the first antenna unit **10** support MHB and UHB in the LTE-4G/NR5G frequency band, and the Wi-Fi 2.4G frequency band, and to make the second antenna unit **20** support the Wi-Fi 5G frequency band and the Wi-Fi 6E frequency band. The second antenna unit **20** can separately support the Wi-Fi frequency band without devices such as a combiner, so as to reduce losses of the second antenna unit **20** and to improve radiation efficiency of the antenna assembly **100**.

(179) The above antenna assembly **100** is disposed in the electronic device **1000**, and the electronic device **1000** as a mobile phone is taken as an example. The present disclosure does not limit specific positions where radiators of the antenna assembly **100** are mounted on the electronic device **1000**. Radiators of the antenna assembly **100** are integrated into the housing **200**, or are disposed on a surface of the housing **200**, or are disposed in the space enclosed by the housing **200**.

(180) As illustrated in FIG. 23, the electronic device **1000** includes a reference ground **GND** and a circuit board **500** (as illustrated in FIG. 2) that are disposed in the housing **200**. The reference ground **GND** includes, but is not limited to, an alloy on the middle plate **410**. The reference ground **GND** includes the first reference ground **GND1**, the second reference ground **GND2**, the third reference ground **GND3**, the fourth reference ground **GND4**, the fifth reference ground **GND5**, the sixth reference ground **GND6**, and the seventh reference ground **GND7**.

(181) The first feeding module **13**, the second feeding module **23**, the first matching module **M1**, and the second matching module **M2** are all disposed on the circuit board **500**.

(182) The first radiator **11** and the second radiator **21** may be integrated into the housing **200**, or may be disposed on the surface of the housing **200**, or may be disposed in the space enclosed by the housing **200**.

(183) Optionally, at least a part of the first radiator **11** and a part of the second radiator **21** are integrated into the frame **210** of the housing **200**. For example, the frame **210** is made of metal. The first radiator **11** and the second radiator **21** are integrated with the frame **210**. The first coupling gap **140** between the first radiator **11** and the second radiator **21** is filled with an insulating material. Certainly, in other embodiments, the first radiator **11** and the second radiator **21** may also be integrated with the back cover **220**. In other words, the first radiator **11** and the second radiator **21** are integrated as a part of the housing **200**.

(184) Optionally, the first radiator **11** and the second radiator **21** are formed on a surface of the frame **210** (for example, an inner surface or an outer surface of the frame **210**). Specifically, the first radiator **11** and the second radiator **21** are basically formed on the inner surface of the frame **210** by, including but not limited to, patch radiators, laser direct structuring (LDS), print direct structuring (PDS), or the like. In this embodiment, a material of the frame **210** may be a non-

conductive material (for the electromagnetic wave signal, an unshielded material is used or a transparent structure is set). Certainly, the first radiator **11** and the second radiator **21** may also be disposed on the surface of the back cover **220**.

(185) Optionally, the first radiator **11** and the second radiator **21** are disposed on a flexible printed circuit board (FPCB), a rigid circuit board, or other carrier boards. The first radiator **11** and the second radiator **21** may be integrated into the FPCB, and the FPCB is attached to the inner surface of the middle frame **420** by an adhesive. In this embodiment, the material of the frame **210** may be the non-conductive material. Certainly, the first radiator **11** and the second radiator **21** may also be disposed on the inner surface of the back cover **220**.

(186) The above is a specific structure of the antenna assembly **100** for detecting the approach of the subject to-be-detected and transmitting antenna signals, and installation positions of each device in the antenna assembly **100** on the electronic device **1000**. Certainly, the antenna assembly **100** may be implemented as one or more antenna assemblies **100** in the present disclosure.

(187) The present disclosure does not limit specific sides of the electronic device **1000** where the antenna assembly **100** may be positioned. The reference ground GND is in the shape of a rectangular plate. The reference ground GND includes multiple sides connected in sequence, and a junction between two adjacent sides is a corner.

(188) The first radiator **11** and the second radiator **21** of the at least one antenna assembly **100** are disposed facing two intersecting sides and the corner; and/or both the first radiator **11** and the second radiator **21** of the at least one antenna assembly **100** are disposed facing one side. Specific examples are illustrated by the following embodiments.

(189) As illustrated in FIG. 23, the reference ground GND includes a first side **61**, a second side **62** disposed opposite to the first side **61**, a third side **63** connected between the first side **61** and the second side **62**, and a fourth side **64** connected between the first side **61** and the second side **62**. A corner **65** is formed at a junction between two adjacent sides. Taking a state where a user holds and uses the electronic device **1000** in portrait mode as a reference, the first side **61** is a top side of the reference ground GND, and the second side **62** is a bottom side of the reference ground GND.

(190) As illustrated in FIG. 23, the frame **210** includes multiple side frames connected end to end. Among multiple side frames of the frame **210**, two adjacent side frames intersect, for example, two adjacent side frames are transitionally connected through a circular arc chamfer. Multiple side frames each includes a top frame **2101**, a bottom frame **2102** disposed opposite to the top frame **2101**, a first side frame **2103** connected between the top frame **2101** and the bottom frame **2102**, and a second side frame **2104** connected between the top frame **2101** and the bottom frame **2102**. When the user holds and uses the electronic device **1000** towards the front of the electronic device **1000**, the top frame **2101** is a side away from the ground, and the bottom frame **2102** is a side close to the ground. A corner portion **2106** is formed at the junction between two adjacent side frames. The top frame **2101** is parallel to and equal to the bottom frame **2102**, the first side frame **2103** is parallel to and equal to the second side frame **2104**, and a length of the first side frame **2103** is greater than a length of the top frame **2101**. The top frame **2101** is disposed facing the first side **61**, the bottom frame **2102** is disposed facing the second side **62**, the first side frame **2103** is disposed facing the third side **63**, and the second side frame **2104** is disposed facing the fourth side **64**.

(191) The antenna assembly **100** is disposed on at least one of the top frame **2101**, the first side frame **2103**, the second side frame **2104**, or the bottom frame **2102**.

(192) As illustrated in FIG. 23, one antenna assembly **100** is provided in one embodiment. The first radiator **11** and the second radiator **21** of the antenna assembly **100** are integrated with the first side frame **2103**. The first coupling gap **140** is close to or defined at the middle of the first side frame **2103**.

(193) As illustrated in FIG. 24, dotted boxes on both sides in FIG. 24 represent holding areas of fingers. When the user holds the electronic device **1000** in landscape mode (for example, playing a game or watching a video in landscape mode), holding areas of user's hands are relatively far away

from the middle of the first side frame **2103**, and the first coupling gap **140** is far away from holding areas of user's hands. When the user's fingers get closer to a strong electric-field point (namely, the first coupling gap **140**) of the antenna assembly **100**, an interference impact on the antenna assembly **100** will be greater. Therefore, the antenna assembly **100** is disposed on the first side frame **2103**, and the first coupling gap **140** is disposed far away from holding areas when the user holds the electronic device **1000** in landscape mode, so that the first coupling gap **140** can be far away from user's fingers when the user holds the electronic device **1000** in landscape mode to play a game or the like, to avoid being blocked. Optionally, a distance between the first coupling gap **140** and the top frame **2101** or and the bottom frame **2102** is greater than or equal to 40 mm, so that the first antenna unit **10** and the second antenna unit **20** still have higher radiation efficiency, and user's experience with the electronic device **1000** can be improved.

(194) FIG. **25** is a schematic graph illustrating the efficiency of the antenna assembly disposed on the side of the electronic device **1000**. As illustrated in FIG. **25**, **S11** illustrates system radiation efficiency of the first antenna unit **10**, **S12** illustrates system radiation efficiency of the second antenna unit **20**, **S13** illustrates overall system efficiency of the first antenna unit **10**, and **S14** illustrates overall system efficiency of the second antenna unit **20**. The antenna unit **100** has a relatively bad radiation environment due to small clearance areas at sides, but it can be seen from reference signs 1 to 6 in FIG. **25** that, the antenna unit **100** still has relatively high efficiency in frequency bands supported by the first resonant mode a to the third resonant mode e (for example, lines over -6 dB in FIG. **25** are taken as reference lines with higher efficiency), so as to satisfy application of the antenna assembly **100** in the electronic device **1000**.

(195) Certainly, as illustrated in FIG. **26**, multiple antenna assemblies **100** are provided in other embodiments, for example, one antenna assembly **100** is disposed at the first side frame **2103** and the other antenna assembly **100** is disposed at the second side frame **2104**. Structures of two antenna assemblies **100** and frequency bands supported by two antenna assemblies **100** can be the same or different. When each of two antenna assemblies **100** has a coupling gap, a coupling gap of one antenna assembly **100** is disposed at or near the middle of the first side frame **2103**, and a coupling gap of the other antenna assembly **100** is disposed at or near the middle of the second side frame **2104**, so as to avoid from blocking the antenna assemblies **100** when the user holds the electronic device **1000** in landscape mode, and to reduce the influence on the efficiency of two antenna assemblies **100**. Meanwhile, two antenna assemblies **100** can be switched to each other, so that the electronic device **1000** can perform communication in more frequency bands or with wider bandwidth, thereby improving communication quality of the electronic device **1000**.

(196) Certainly, as illustrated in FIG. **27**, in other embodiments, the first radiator **11** and the second radiator **21** of the antenna assembly **100** may also be integrated at the top frame **2101** and closer to the corner, or may also be integrated at any position of the electronic device **1000**.

(197) It can be seen from the described embodiments that, in the present disclosure, the electronic device **1000** has higher radiation performance in landscape mode, and LTE-4G/NR5G/Wi-Fi frequency bands exist simultaneously.

(198) The antenna assembly **100** provided in a second type of embodiment of the present disclosure is described below with reference to accompanying drawings. The antenna assembly **100** provided in this embodiment includes the antenna assembly **100** provided in the first type of embodiment. The difference lies in that, as illustrated in FIG. **28** and FIG. **29**, the antenna assembly **100** provided in this embodiment further includes a third radiator **31**, a third feeding module **33**, and a third matching module **M3**. To facilitate functional division of different parts of the antenna assembly **100** for subsequent descriptions, the third radiator **31**, the third feeding module **33**, and the third matching module **M3** are defined as a third antenna unit **30**. The third radiator **31** includes a third ground end **311**, a first free end **312**, and a third feeding point C between the third ground end **311** and the first free end **312**. The third ground end **311** is spaced apart from the first ground end **111** or is connected to the first ground end **111** via a conductor. The third ground end **311** is electrically

connected to the fifth reference ground GND5. The third feeding module 33 is electrically connected to the third feeding point C. The third matching module M3 is electrically connected between the third feeding point C and the third feeding module 33. Certainly, in other embodiments, positions of the first antenna unit 10 and the second antenna unit 20 may be interchanged.

(199) Optionally, references are made to FIG. 28 and FIG. 30, the first radiator 11 and the second radiator 12 are disposed on the first side frame 2103, so that the first coupling gap 140 is far away from the user's hands to increase the radiation efficiency of the electronic device 1000 when the electronic device 1000 is used in landscape mode. The third radiator 31 is disposed at the corner portion 2106 between the first side frame 2103 and the top frame 2101. The corner portion 2106 has a relatively good clearance area, and the corner portion 2106 can more easily excite higher reference ground current, so that radiation efficiency of the third antenna unit 30 can be improved. The present disclosure does not limit frequency bands supported by the third antenna unit 30. Optionally, the third antenna unit 30 may support (GPS-L1)+(WI-FI 2.4G)+(LTE/NR-MHB+UHB). The structure and size of the third antenna unit 30 may refer to the structure and size of the first antenna unit 10.

(200) Optionally, as illustrated in FIG. 30, a conductor is additionally disposed between the first ground end 111 and the third ground end 311 to connect the first radiator 11 and the third radiator 31. Further, when the first radiator 11, the second radiator 21, and the third radiator 31 are all integrated on the frame 210, the first radiator 11 may be integrated with the third radiator 31, that is, a part of the frame 210 is used as the first radiator 11, and a part of the frame 210 is used as the third radiator 31. In other words, the first radiator 11 and the third radiator 31 form a sharing radiator 32.

(201) Optionally, as illustrated in FIG. 31, the antenna assembly 100 further includes a fourth matching module M4. One end of the fourth matching module M4 is electrically connected to the first ground end 111, and the other end of the fourth matching module M4 is electrically connected to the first reference ground GND1. The fourth matching module M4 has low impedance on frequency bands supported by resonant modes (for example, the first resonant mode a) whose resonant current needs to return to the ground at the first ground end 111 in the first antenna unit 10 and the second antenna unit 20. For example, the fourth matching module M4 is an inductor, so that resonant current of resonant modes that needs to return to the ground at the first ground end 111 can return to the ground.

(202) Optionally, as illustrated in FIG. 31, the antenna assembly 100 further includes a fifth matching module M5. One end of the fifth matching module M5 is electrically connected to the third ground end 311, and the other end of the fifth matching module M5 is electrically connected to the fifth reference ground GND5. The fifth matching module M5 has low impedance on frequency bands supported by resonant modes (for example, the third resonant mode e) whose resonant current needs to return to the ground at the first ground end 111 in the third antenna unit 30. For example, the fifth matching module M5 is an inductor, so that resonant current of resonant modes that needs to return to the ground at the first ground end 111 in the third antenna unit 30 can return to the ground.

(203) In this embodiment, the antenna assembly 100 includes the fourth matching module M4 and the fifth matching module M5. In other embodiments, the antenna assembly 100 may only include the fourth matching module M4, or only include the fifth matching module M5.

(204) As illustrated in FIG. 32, the first radiator 11 is electrically connected to the third radiator 31, and combination of the first radiator 11 and the third radiator 31 is defined as the sharing radiator 32. When the sharing radiator 32 is used as a sensing electrode for detecting the approach of the subject to-be-detected (for example, human's head, hands, or the like), the fourth matching module M4 includes a capacitive device directly electrically connected to the first ground end 111, or a capacitive device disposed between the fourth matching module M4 and the first ground end 111.

The fifth matching module M5 includes a capacitive device directly electrically connected to the third ground end **311**, or a capacitive device disposed between the fifth matching module M5 and the third ground end **311**. The first matching module M1 includes a capacitive device directly electrically connected to the first feeding point A, or a capacitive device disposed between the first feeding point A and the first matching module M1. The third matching module M3 includes a capacitive device directly electrically connected to the third feeding point C, or a capacitive device disposed between the third feeding point C and the third matching module M3. The capacitive device, such as a capacitor or a circuit including a capacitor, makes the sharing radiator **32** in a “floating state” relative to direct current, so as to avoid an antenna signal from being interrupted by a sensing signal generated in response to the approach of the subject to-be-detected.

(205) Optionally, each of the fourth matching module M4 and the fifth matching module M5 includes a switch selection circuit, for example, an inductor branch and a capacitor branch that are connected in parallel. When the sharing radiator **32** needs to detect approach of a human body, a switch can be controlled to conduct the capacitor to the first ground end **111**. When the sharing radiator **32** does not need to detect the approach of the human body, the switch can be controlled to conduct the inductor to the first ground end **111**.

(206) Optionally, each of the fourth matching module M4 and the fifth matching module M5 further includes a capacitor branch and an inductor branch. The capacitor is electrically connected between the first ground end **111** and the inductor, and the capacitor enables the sharing radiator **32** to be in the “floating state” relative to the DC. Both the capacitor branch and the inductor branch have low impedance on frequency bands corresponding to resonant modes whose resonant current needs to return to the ground at the first ground end **111** and the third ground end **311**.

(207) As illustrated in FIG. **32**, each of the fourth matching module M4 and the fifth matching module M5 is configured to block direct current and match, so that the sharing radiator **32** may be used for detecting a specific absorption rate (SAR). Optionally, a DC blocking capacitor C31 is additionally disposed before the first matching module M1, a DC blocking capacitor C32 is additionally disposed before the fourth matching module M4, a DC blocking capacitor C33 is additionally disposed before the fifth matching module M5, and a DC blocking capacitor C34 is additionally disposed before the third matching module M3, where all of DC blocking capacitors has a capacitance, for example, 22 pF, and has no impact on the antenna signal). If the first matching module M1 already includes a DC blocking capacitor, the DC blocking capacitor C31 does not need to be additionally disposed before the first matching module M1. If the fourth matching module M4 already includes a DC blocking capacitor, the DC blocking capacitor C32 does not need to be additionally disposed before the fourth matching module M4. If the fifth matching module M5 already includes a DC blocking capacitor, the DC blocking capacitor C33 does not need to be additionally disposed before the fifth matching module M5. If the third matching module M3 already includes a DC blocking capacitor, the DC blocking capacitor C34 does not need to be additionally disposed before the third matching module M3. In this case, the sharing radiator **32** is floating for the sensing signal, and a proximity sensor needs a floating metal body to sense capacitor changes caused by the approach of the human body, so as to achieve the purpose of detection. A detecting circuit is additionally disposed before the C33, and an inductor L is additionally disposed in the detecting circuit to block relatively high frequencies (for example, an inductance is 82 nH), so that the antenna is basically not affected. The detecting circuit may also be placed before C31/C32/C34 or at any position of an antenna radiator. Detecting the human body with the antenna radiator can be used to determine the approach of the human body, so as to intelligently reduce the SAR.

(208) The above are embodiments in which the sharing radiator **32** is used as the sensing electrode, and in other embodiments, the second radiator **21** can also be used as the sensing electrode, which will be described in detail below by taking the antenna assembly **100** provided in a third type of embodiment as an example.

(209) A specific structure of the antenna assembly **100** provided in the third type of embodiment provided in the present disclosure is illustrated below with reference to accompanying drawings.

(210) Optionally, as illustrated in FIG. **33**, the antenna assembly **100** further includes a fourth radiator **41**, a fourth feeding module **43**, and a sixth matching module **M6**. To facilitate the functional division of different parts of the antenna assembly **100** for subsequent descriptions, the fourth radiator **41**, the fourth feeding module **43**, and the sixth matching module **M6** are defined as a fourth antenna unit **40**. The fourth radiator **41** includes a fourth ground end **412**, a second free end **411**, and a fourth feeding point D between the fourth ground end **412** and the second free end **411**. A second coupling gap **150** is defined between the second free end **411** and the first free end **312**. The third radiator **31** is coupled with the fourth radiator **41** through the second coupling gap **150**. The fourth ground end **412** is electrically connected to the sixth reference ground **GND6**, the fourth feeding module **43** is electrically connected to the fourth feeding point D, and the sixth matching module **M6** is electrically connected between the fourth feeding point D and the fourth feeding module **43**.

(211) In this embodiment, the third radiator **31** and the fourth radiator **41** are coupled to each other, so that excitation current flowing from the fourth feeding module **43** can not only resonate on the fourth radiator **41**, but also resonate on the third radiator **31**; and excitation current flowing from the third feeding module **33** can not only resonate on the third radiator **31**, but also resonate on the fourth radiator **41**. In this way, resonant modes generated by the third antenna unit **30** and the fourth antenna unit **40** are increased, and bandwidth supported by the third antenna unit **30** and the fourth antenna unit **40** and the number of frequency bands supported by the third antenna unit **30** and the fourth antenna unit **40** are increased.

(212) Optionally, the third radiator **31** and the fourth radiator **41** may jointly serve as radiators of the third antenna unit **30**, and the third radiator **31** and the fourth radiator **41** may also jointly serve as radiators of the fourth antenna unit **40**. In this way, the third antenna unit **30** and the fourth antenna unit **40** can form a co-aperture antenna to realize the reuse of radiators, and the overlapping length of radiators of the third antenna unit **30** and the fourth antenna unit **40** is reduced while the number of resonant modes is increasing.

(213) Optionally, the first radiator **11** is electrically connected to the third radiator **31**. The first antenna unit **10**, the second antenna unit **20**, the third antenna unit **30**, and the fourth antenna unit **40** form the co-aperture antenna.

(214) In the antenna assembly **100** provided in this embodiment, resonant modes and resonant current of the first antenna unit **10** and the second antenna unit **20** may be referred to the antenna assembly **100** provided in the first type of embodiment, and details are not repeatedly described herein.

(215) As illustrated in FIG. **34**, which is a schematic graph illustrating S-parameters of the first antenna unit **10**. As illustrated in FIG. **34**, the resonant frequency of the first resonant mode a is 1.88 GHz, the resonant frequency of the second resonant mode b is 2.6323 GHz, and the resonant frequency of the fourth resonant mode c is 3.5707 GHz.

(216) As illustrated in FIG. **35**, which is a schematic graph illustrating S-parameters of the second antenna unit **20**. As illustrated in FIG. **34**, the resonant frequency of the fifth resonant mode d is 5.4751 GHz, and the resonant frequency of the third resonant mode e is 5.9643 GHz.

(217) Resonant modes and resonant current of the third antenna unit **30** and the fourth antenna unit **40** will be described below with reference to accompanying drawings.

(218) As illustrated in FIG. **36**, the third antenna unit **30** is configured to support a sixth resonant mode f, a seventh resonant mode g, and an eighth resonant mode h.

(219) As illustrated in FIG. **37**, sixth-resonant-current density of the sixth resonant mode f is mainly distributed between the third ground end **311** and the first free end **312**, and sixth resonant current flows from the third ground end **311** to the first free end **312**, or flows from the first free end **312** to the third ground end **311**. It can be understood that, current density of the sixth resonant

mode f is mainly distributed as described above, and a small amount of current density of the sixth resonant mode f is distributed on the fourth radiator **41**.

(220) The sixth resonant mode f is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module **33** resonating on the third radiator **31**. Specifically, a length of the third radiator **31** is designed to be about $\frac{1}{8}$ times to $\frac{1}{4}$ times a wavelength of the excitation current, in the medium, flowing from the third feeding module **33** to the third radiator **31**, so that the sixth resonant mode f is excited on the third radiator **31**, and higher radiation efficiency can be achieved in a frequency band supported by the sixth resonant mode f.

(221) Generally, the length of the third radiator **31** is designed to be about $\frac{1}{4}$ times the wavelength of the excitation current, in the medium, flowing from the third feeding module **33** to the third radiator **31**, so that the higher radiation efficiency is easily excited at a sixth resonant frequency point (a resonant frequency of the sixth resonant mode f). Further, the capacitive matching circuit connected to the ground is disposed on a path of the sixth resonant current, so as to feed the sixth resonant current to the third radiator **31** through capacitive coupling. The capacitive loading may enable the resonant frequency of the sixth resonant mode f to be shifted towards lower frequencies, so that the higher-efficiency resonance is not just generated when the length of the third radiator **31** is about $\frac{1}{4}$ wavelength, but also generated when the length of the third radiator **31** is in the range of the $\frac{1}{8}$ to $\frac{1}{4}$ wavelength. Therefore, while the resonance is generated at the sixth resonant frequency point, the length of the third radiator **31** can also be shortened, for example, shortened to $\frac{1}{8}$ times a wavelength corresponding to the sixth resonant frequency point, so as to further reduce the length of the third radiator **31** and to reduce the overlapping length of the antenna assembly **100**.

(222) As illustrated in FIG. **38**, seventh-resonant-current density of the seventh resonant mode g is mainly distributed between the third feeding point C and the first free end **312**, and seventh resonant current flows from the third feeding point C to the first free end **312**, or flows from the first free end **312** to the third feeding point C. It can be understood that, current density of the seventh resonant mode g is mainly distributed as described above, and a small amount of current density of the seventh resonant mode g is distributed between the third ground end **311** to the third feeding point C and distributed on the fourth radiator **41**.

(223) The seventh resonant mode g is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module **33** resonating between the third feeding point C and the first free end **312**. Specifically, a length from the third feeding point C to the first free end **312** is designed to be about $\frac{1}{8}$ times to $\frac{1}{4}$ times a wavelength of the excitation current, in the medium, flowing from the third feeding module **33** to the third feeding point C and the first free end **312**, so that the seventh resonant mode g is excited between the third feeding point C and the first free end **312**, and higher radiation efficiency can be achieved in a frequency band supported by the seventh resonant mode g.

(224) Generally, the length from the third feeding point C to the first free end **312** is designed to be about $\frac{1}{4}$ times the wavelength of the excitation current, in the medium, between the third feeding point C and the first free end **312** flowing from the third feeding module **33**, so that the higher radiation efficiency is easily excited at a seventh resonant frequency point (a resonant frequency of the seventh resonant mode g). Further, the capacitive matching circuit connected to the ground is disposed on a path of the seventh resonant current, so as to feed the seventh resonant current between the third feeding point C and the first free end **312** through capacitive coupling. The capacitive loading may enable the resonant frequency of the seventh resonant mode g to be shifted towards lower frequencies, so that the higher-efficiency resonance is not just generated when the length from the third feeding point C to the first free end **312** is about $\frac{1}{4}$ wavelength, but also generated when the length from the third feeding point C to the first free end **312** is in the range of the $\frac{1}{8}$ to $\frac{1}{4}$ wavelength. Therefore, while the resonance is generated at the seventh resonant frequency point, the length from the third feeding point C to the first free end **312** can also be shortened, for example, shortened to $\frac{1}{8}$ times a wavelength corresponding to the seventh resonant frequency point, so as to further reduce the length of the third radiator **31** and to reduce the

overlapping length of the antenna assembly **100**.

(225) As illustrated in FIG. **39**, eighth-resonant-current density of the eighth resonant mode **h** is mainly distributed between the third feeding point **C** and the fourth ground end **412**. A direction of current flowing between the third feeding point **C** to the first free end **312** is the same as a direction of current between the second free end **411** to the fourth ground end **412**. Eighth resonant current flows from the third feeding point **C** to the first free end **312**, through the second coupling gap **150** to the second free end **411**, and from the second free end **411** to the fourth ground end **412**; or the eighth resonant current flows from the fourth ground end **412** to the second free end **411**, through the second coupling gap **150** to the first free end **312**, and from the first free end **312** to the third feeding point **C**. It can be understood that, current density of the eighth resonant mode **h** is mainly distributed as described above, and a small amount of current density of the eighth resonant mode **h** is distributed between the third ground end **311** and the third feeding point **C**.

(226) The eighth resonant mode **h** is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module **33** resonating between the third feeding point **C** and the first free end **312**, and is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module **33** resonating between the second free end **411** and the fourth ground end **412**.

(227) Generally, the length from the third feeding point **C** to the first free end **312** is designed to be about $\frac{1}{4}$ times a wavelength of the excitation current, in the medium, between the third feeding point **C** and the first free end **312** flowing from the third feeding module **33**, and a length from the second free end **411** to the fourth ground end **412** is designed to be about $\frac{1}{4}$ times a wavelength of the excitation current, in the medium, between the second free end **411** and the fourth ground end **412** flowing from the third feeding module **33**, so that the higher radiation efficiency is easily excited at an eighth resonant frequency point (a resonant frequency of the eighth resonant mode **h**). Further, the capacitive matching circuit connected to the ground is disposed between the third feeding point **C** and the first free end **312**, and the capacitive matching circuit connected to the ground is disposed between the second free end **411** and the fourth ground end **412**. The capacitive loading may enable the resonant frequency of the eighth resonant mode **h** to be shifted towards lower frequencies, so that the higher-efficiency resonance is not just generated when the length from the third feeding point **C** to the first free end **312** is about $\frac{1}{4}$ wavelength, but also generated when the length from the third feeding point **C** to the first free end **312** is in the range of the $\frac{1}{8}$ to $\frac{1}{4}$ wavelength; and the higher-efficiency resonance is not just generated when the length from the second free end **411** to the fourth ground end **412** is about $\frac{1}{4}$ wavelength, but also generated when the length from the second free end **411** to the fourth ground end **412** is in the range of the $\frac{1}{8}$ to $\frac{1}{4}$ wavelength. Therefore, while the resonance is generated at the eighth resonant frequency point, the length from the third feeding point **C** to the first free end **312** and the length of the fourth radiator **41** can also be shortened, for example, shortened to $\frac{1}{8}$ times a wavelength corresponding to the eighth resonant frequency point, so as to further reduce the length of the third radiator **31** and the fourth radiator **41**, and to reduce the overlapping length of the antenna assembly **100**.

(228) As illustrated in FIG. **40**, the fourth antenna unit **40** is configured to support a ninth resonant mode **i**, a tenth resonant mode **j**, an eleventh resonant mode **k**, and a twelfth resonant mode **p**.

(229) As illustrated in FIG. **39**, the ninth resonant mode **i** is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module **33** resonating between the third feeding point **C** and the first free end **312**, and is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module **33** resonating between the second free end **411** and the fourth ground end **412**. For ninth-resonant-current density distribution of the ninth resonant mode **i**, reference may be made to the eighth resonant mode **h** distribution and the eighth-resonant-current-density distribution, and details are not repeatedly described herein.

(230) As illustrated in FIG. **41**, tenth-resonant-current density of the tenth resonant mode **j** is mainly distributed between the third feeding point **C** and the fourth ground end **412**. A direction of current between the third feeding point **C** and the first free end **312** and a direction of current

between the second free end **411** and the fourth ground end **412** are opposite. A part of tenth resonant current flows from the third feeding point C to the first free end **312**, and the other part of the tenth resonant current flows from the fourth ground end **412** to the second free end **411**; or a part of the tenth resonant current flows from the first free end **312** to the third feeding point C, and the other part of the tenth resonant current flows from the second free end **411** to the fourth ground end **412**. It can be understood that, current density of the tenth resonant mode j is mainly distributed as described above, and a small amount of current density of the tenth resonant mode j is distributed between the third grounding terminal **311** and the third feeding point C.

(231) The tenth resonant mode j is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module **33** resonating between the third feeding point C and the first free end **312**, and a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module **33** resonating between the second free end **411** and the fourth ground end **412**.

(232) Generally, the length of the third feeding point C to the first free end **312** is designed to be about $\frac{1}{4}$ times a wavelength of the excitation current, in the medium, between the third feeding point C and the first free end **312** flowing from the third feeding module **33**, and the length from the second free end **411** to the fourth ground end **412** is designed to be about $\frac{1}{4}$ times a wavelength of the excitation current, in the medium, between the second free end **411** and the fourth ground end **412** flowing from the third feeding module **33**, so that the higher radiation efficiency is easily excited at a tenth resonant frequency point (a resonant frequency of the tenth resonant mode j). Further, the capacitive matching circuit connected to the ground is disposed between the third feeding point C and the first free end **312**, and the capacitive matching circuit connected to the ground is disposed between the second free end **411** and the fourth ground end **412**. The capacitive loading may enable the resonant frequency of the eighth resonant mode h to be shifted towards lower frequencies, so that the higher-efficiency resonance is not just generated when the length from the third feeding point C to the first free end **312** is about $\frac{1}{4}$ wavelength, but also generated when the length from the third feeding point C to the first free end **312** is in the range of the $\frac{1}{8}$ to $\frac{1}{4}$ wavelength; and the higher-efficiency resonance is not just generated when the length from the second free end **411** to the fourth ground end **412** is about $\frac{1}{4}$ wavelength, but also generated when the length from the second free end **411** to the fourth ground end **412** is in the range of the $\frac{1}{8}$ to $\frac{1}{4}$ wavelength. Therefore, while the resonance is generated at the tenth resonant frequency point, the length from the third feeding point C to the first free end **312** and the length of the fourth radiator **41** can also be shortened, for example, shortened to $\frac{1}{8}$ times a wavelength corresponding to the tenth resonant frequency point, so as to further reduce the length of the third radiator **31** and the fourth radiator **41**, and to reduce the overlapping length of the antenna assembly **100**.

(233) As illustrated in FIG. **42**, eleventh-resonant-current density of the eleventh resonant mode k is mainly distributed between the fourth feeding point D and the second free end **411**, and eleventh resonant current flows from the fourth feeding point D to the second free end **411**, or flows from the second free end **411** to the fourth feeding point D. It can be understood that, current density of the eleventh resonant mode k is mainly distributed as described above, and a small amount of current density of the eleventh resonant mode k is distributed between the fourth feeding point D and the second free end **411** and distributed on the first radiator **11**.

(234) Specifically, the eleventh resonant mode k is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current (namely, the eleventh resonant current) of the fourth feeding module **43** resonating between the fourth feeding point D and the second free end **411**. Specifically, a length from the fourth feeding point D to the second free end **411** is designed to be about $\frac{1}{8}$ times to $\frac{1}{4}$ times a wavelength of the excitation current, in the medium, transmitted from the fourth feeding module **43** to the fourth radiator **41**, so as to excite the eleventh resonant mode k between the fourth feeding point D and the second free end **411**.

(235) Generally, the length from the fourth feeding point D to the second free end **411** is designed to be about $\frac{1}{4}$ times the wavelength of the excitation current, in the medium, flowing from the

fourth feeding module **43**, so that the higher radiation efficiency is easily excited at an eleventh resonant frequency point (namely, a resonant frequency of the eleventh resonant mode **k**). Further, the capacitive matching circuit connected to the ground is disposed on a path of the eleventh resonant current, so as to feed the eleventh resonant current to the fourth radiator **41** through capacitive coupling. The capacitive loading may enable the resonant frequency of the eleventh resonant mode **k** to be shifted towards lower frequencies, so that the higher-efficiency resonance is not just generated when the length from the fourth feeding point **D** to the second free end **411** is about $\frac{1}{4}$ wavelength, but also generated when the length from the fourth feeding point **D** to the second free end **411** is in the range of the $\frac{1}{8}$ to $\frac{1}{4}$ wavelength. Therefore, while the resonance is generated at the eleventh resonant frequency point, the length from the fourth feeding point **D** to the second free end **411** can also be shortened, for example, shortened to $\frac{1}{8}$ times a wavelength corresponding to the eleventh resonant frequency point, so as to further reduce the length from the fourth feeding point **D** to the second free end **411** and to reduce the overlapping length of the antenna assembly **100**.

(236) As illustrated in FIG. **43**, twelfth-resonant-current density of the twelfth resonant mode **p** is mainly distributed between the third ground end **311** and the first free end **312**, and twelfth resonant current flows from the third ground end **311** to the first free end **312**, or flows from the first free end **312** to the third ground end **311**. It can be understood that, current density of the twelfth resonant mode **p** is mainly distributed as described above, and a small amount of current density of the twelfth resonant mode **p** is distributed between the second free end **411** and the fourth ground end **412**.

(237) The twelfth resonant mode **p** is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the fourth feeding module **43** resonating on the third radiator **31**. Specifically, the length of the third radiator **31** is designed to be about $\frac{1}{8}$ times to $\frac{1}{4}$ times a wavelength of the excitation current, in the medium, flowing from the fourth feeding module **43** to the third radiator **31**, so as to excite the twelfth resonant mode **p** on the third radiator **31**.

(238) It should be noted that, the sixth resonant mode **f** to the twelfth resonant mode **p** are described above from the perspective of the wavelength mode, which can be better understood and explained and illustrates main characteristics of each resonant mode so that each resonant mode is easily distinguished. However, when each resonant mode operates, the third antenna unit **30** and the fourth antenna unit **40** are not separate but are coupled to each other, and current also flows to each other through the coupling.

(239) The third radiator **31** supports, under excitation of the third feeding module **33** (namely, the third antenna unit **30**), the GPS frequency band, the Wi-Fi 2.4G frequency band, the LTE-4G MHB frequency band, the NR-5G MHB frequency band, the LTE-4G UHB frequency band, and the NR-5G UHB frequency band. For example, the resonant frequency of the sixth resonant mode **f** is 1.5766 GHz, the resonant frequency of the seventh resonant mode **g** is 2.4667 GHz, and the resonant frequency of the eighth resonant mode **h** is 2.9773 GHz. The third antenna unit **30** covers (GPS-L1)+(Wi-Fi 2.4G)+(LTE/NR-MHB+UHF). In other embodiments, the third antenna unit **30** covers (GPS-L1)+(Wi-Fi 2.4G)+(LTE/NR-MHB+UHF)+N77/N78.

(240) The fourth radiator **41** supports, under excitation of the fourth feeding module **43** (namely, the fourth antenna unit **40**), the N77 frequency band, the N78 frequency band, the Wi-Fi 5G frequency band, and the Wi-Fi 6E frequency band. For example, the resonant frequency of the ninth resonant mode **i** is 2.998 GHz, the resonant frequency of the tenth resonant mode **j** is 3.6742 GHz, the resonant frequency of the eleventh resonant mode **k** is 5.5096 GHz, and the resonant frequency of the twelfth resonant mode **p** is 6.5722 GHz. The fourth antenna unit **40** covers the N77/N78+Wi-Fi 5G frequency band and the Wi-Fi 6E frequency band. Certainly, in other embodiments, the fourth antenna unit **40** covers the Wi-Fi 5G frequency band and the Wi-Fi 6E frequency band.

(241) The arrangement of the antenna assembly **100** in the electronic device **1000** provided in this

embodiment includes, but is not limited to, the following embodiments. The first radiator **11** and the second radiator **12** are both disposed on the first side frame **2103**, and the first coupling gap **140** is defined at the middle of the first side frame **2103**. A part of the third radiator **31** is disposed on the first side frame **2103**, and the other part of the third radiator **31** is disposed on the top frame **2101**. The fourth radiator **41** is disposed on the top frame **2101**, and the second coupling gap **150** is defined on the top frame **2101**.

(242) When the user uses the electronic device **1000** in landscape mode, the first coupling gap **140** is away from holding areas of the user's hands, so that the first antenna unit **10** and the second antenna unit **20** can have higher radiation efficiency to receive/transmit signals in frequency bands of (GPS-L1)+(WI-FI 2.4G)+(LTE/NR-MHB+UHF)+N77/N78+Wi-Fi 5G/Wi-Fi 6E, so as to improve user's experience in landscape mode. When the user uses the electronic device **1000** in portrait mode, the second coupling gap **150** is away from holding areas of the user's hands, so that the third antenna unit **30** and the fourth antenna unit **40** can have higher radiation efficiency to receive/transmit signals in frequency bands of (GPS-L1)+(WI-FI 2.4G)+(LTE/NR-MHB+UHF)+N77/N78+Wi-Fi 5G/Wi-Fi 6E frequency bands, so as to improve user's experience in portrait mode. It can be seen from the above that the antenna assembly **100** provided by the present embodiment can effectively cover frequency bands of (GPS-L1)+(WI-FI 2.4G)+(LTE/NR-MHB+UHB)+N77/N78+Wi-Fi 5G/Wi-Fi 6E both in landscape mode and in portrait mode.

(243) Optionally, the electronic device **1000** further includes at least one key portion **440**, where the at least one key portion **440** is disposed between the second ground end **212** and the second feeding point B, and/or the at least one key portion **440** is disposed between the first ground end **111** and the third ground end **311**.

(244) As illustrated in FIG. **44**, the reference ground GND is an alloy device of the middle plate **410**. When the antenna assembly **100** is disposed in the electronic device **1000**, each device on the antenna assembly **100** is spaced apart from each other, and other devices in the electronic device **1000** may be disposed in space between devices on the antenna assembly **100**. For example, a key circuit board **430** may be disposed in space between the second reference ground GND2 and the second feeding module **23**, and the key portion **440** may be disposed on the second radiator **21**. The key portion **440** includes, but is not limited to, a power key, a volume key, a mute key, or the like. In addition, the key circuit board **430** may also be disposed in space between the fourth matching module M4 and the fifth matching module M5, and the key portion **440** may be disposed on the middle frame between the first ground end **111** and the third ground end **311**. Other devices in the electronic device **1000** are disposed in the space between devices on the antenna assembly **100**, so as to promote compactness of the structure of the electronic device **1000** with the antenna assembly **100** to facilitate the overlapping length of the electronic device **1000**.

(245) It can be understood that, the first matching module M1 to the fifth matching module M5 may be provided with devices in any one of the embodiments in FIG. **11** to FIG. **18**, and may also include an adjustable device such as a switching circuit, a variable capacitor, or the like.

(246) As illustrated in FIG. **45**, which are curves of S-parameters of the first antenna unit **10** to the fourth antenna unit **40**. It can be seen from the curves of S-parameters of total power of the antenna assembly **100** in FIG. **45** that, the antenna assembly **100** can better cover frequency bands of (GPS-L1)+(WI-FI 2.4G)+(LTE/NR-MHB+UHB)+N77/N78+Wi-Fi 5G/Wi-Fi 6E, and has a good impedance bandwidth.

(247) As illustrated in FIG. **46**, which are curves of isolation between antenna units of the antenna assembly **100**. As illustrated in FIG. **46**, S2,1 curve illustrates isolation between the first antenna unit **10** and the second antenna unit **20**, S3,1 curve illustrates isolation between the third antenna unit **30** and the first antenna unit **10**, S3,2 curve illustrates isolation between the third antenna unit **30** and the second antenna unit **20**, S4,1 curve illustrates isolation between the fourth antenna unit **40** and the first antenna unit **10**, S4,2 curve illustrates isolation between the fourth antenna unit **40** and the second antenna unit **20**, and S4,3 curve illustrates isolation between the fourth antenna unit

40 and the third antenna unit **30**. Taking the S parameter as -13 dB as a reference, it can be seen from FIG. **46** that, the curve of isolation between each antenna unit is mostly set below -13 dB, which indicates that each antenna unit has better isolation with one other in the antenna assembly **100**.

(248) The function of the antenna assembly **100** provided in any one of the above embodiments will be further described below with reference to accompanying drawings. For example, the antenna assembly **100** can also detect the approach of the subject to-be-detected while receiving/transmitting antenna signals. The subject to-be-detected includes, but is not limited to, human's head, hands, or the like. It can be understood that, radiators are made of a conductive material, and can also be used as a sensing electrode for proximity signals while serving as an antenna-signal-receiving/transmitting port. The antenna assembly **100** provided in the present disclosure has dual functions of receiving/transmitting the electromagnetic wave signal and sensing proximity, and has a small volume. When the antenna assembly **100** is applicable to the electronic device **1000**, the electronic device **1000** can be ensured to have a communication function and a proximity detecting function and to have a small size.

(249) Specifically, as illustrated in FIG. **47**, the antenna assembly **100** further includes a DC blocking component **70**, a filtering component **50**, a detecting component **80**, and a controller (not illustrated in FIG. **47**).

(250) The following describes examples of connection manners of the DC blocking component **70** and the filtering component **50** with reference to the antenna assembly **100** provided in the third type of embodiment.

(251) As illustrated in FIG. **47**, in a first type of embodiment of proximity detecting, the sharing radiator **32** is used as the sensing electrode for detecting the approach of the subject to-be-detected. The DC blocking component **70** is electrically connected between the first feeding point A and the first matching module M1, between the first ground end **111** and the first reference ground GND1 (when the fourth matching module M4 is disposed between the first ground end **111** and the first reference ground GND1, the DC blocking component **70** is disposed between the first ground end **111** and the fourth matching module M4), between the third ground end **311** and the fifth reference ground GND5 (when the fifth matching module M5 is disposed between the third ground end **311** and the fifth reference ground GND5, the DC blocking component **70** is disposed between the third ground end **311** and the fifth matching module M5), and between the third feeding point C and the third matching module M3. The DC blocking component **70** is configured to block direct current generated by the first matching module M1, the first reference ground GND1, the fifth reference ground GND5, and the third matching module M3, so as to support a human-body-detecting function and to improve accuracy for detecting the approach of the human body to the antenna assembly **100**.

(252) One end of the filtering component **50** is electrically connected to one end of the DC blocking component **70** close to the sharing radiator **32**, or electrically connected to any position of the sharing radiator **32**. The filtering component **50** is configured to block an RF signal received/transmitted by the sharing radiator **32** and a sensing signal generated by the sharing radiator **32** in response to the approach of the subject to-be-detected, so that the RF signal received/transmitted by the first radiator **11** does not affect accuracy for detecting the sensing signal by the detecting component **80**.

(253) Specifically, as illustrated in FIG. **47**, the DC blocking component **70** includes a first sub-isolator **71**, a second sub-isolator **72**, a third sub-isolator **73**, and a fourth sub-isolator **74**. The first sub-isolator **71** is electrically connected between the first ground end **111** and the first reference ground GND1 (specifically, between the first ground end **111** and the fourth matching module M4). The second sub-isolator **72** is electrically connected between the first feeding point A and the first matching module M1. The third sub-isolator **73** is electrically connected between the third ground end **311** and the fifth reference ground GND5 (specifically, between the third ground end **311** and

the fifth matching module M5). The fourth sub-isolator **74** is electrically connected between the third feeding point C and the third matching module M3.

(254) The DC blocking component **70** is disposed so that the sensing signal generated by the sharing radiator **32** in response to approach of the subject to-be-tested does not affect the antenna signal received/transmitted by the antenna assembly **100**. Specifically, each of the first sub-isolator **71**, the second sub-isolator **72**, the third sub-isolator **73**, and the fourth sub-isolator **74** is a capacitive device. For example, each of the first sub-isolator **71**, the second sub-isolator **72**, the third sub-isolator **73**, and the fourth sub-isolator **74** is a capacitor, and each of the first sub-isolator **71**, the second sub-isolator **72**, the third sub-isolator **73**, and the fourth sub-isolator **74** has low impedance to ground on the RF signal supported by the antenna assembly **100**. For example, value of the first sub-isolator **71**, value of the second sub-isolator **72**, value of the third sub-isolator **73**, and value of the fourth sub-isolator **74** include, but are not limited to, 47 pF, 22 pF, or the like. The first sub-isolator **71** blocks direct current of the fourth matching module M4, the second sub-isolator **72** blocks direct current of the first matching module M1, the third sub-isolator **73** blocks direct current of the fifth matching module M5, and the fourth sub-isolator **74** blocks direct current of the third matching module M3. In this way, the human-body-detecting function can be supported, and accuracy for detecting the approach of the human body to the antenna assembly **100** can be improved. In other words, the DC blocking component **70** makes the sharing radiator **32** in the “floating” state relative to the direct current.

(255) Specifically, the filtering component **50** is electrically connected between the first sub-isolator **71** and the first ground end **111**, or is electrically connected between the second sub-isolator **72** and the first feeding point A, or is electrically connected between the third sub-isolator **73** and the third ground end **311**, or is electrically connected between the fourth sub-isolator **74** and the third feeding point c, or is electrically connected to any position of the sharing radiator **32**. The filtering component **50** includes an inductive device, or the filtering component **50** is an inductive device. For example, the filtering component **50** is an inductor. The filtering component **50** has a high impedance on the RF signal supported by the antenna assembly **100**, and inductance of the inductor is, for example, 82 nH.

(256) The above-mentioned DC blocking component **70** and filtering component **50** realize that the sensing signal and the RF signal can work at the same time but do not interfere with each other.

(257) The detecting component **80** is electrically connected to the other end of the filtering component **50**. The detecting component **80** is configured to detect the magnitude of a sensing signal generated by a radiator. Optionally, the detecting component **80** may be a device for detecting a current signal, a voltage signal, or an inductor signal, for example, a micro galvanometer, a micro current transformer, a current comparator, a voltage comparator, or the like.

(258) When human skin surface is close to the sharing radiator **32**, the human skin surface and the sharing radiator **32** may be equivalent to two electrode plates of a capacitor. When human's head approaches, the sharing radiator **32** may sense a change in electrical charges brought by the human's head. The filtering component **50** is electrically connected to the sharing radiator **32**. The change in electrical charges forms a sensing signal, and the sensing signal is transmitted to the detecting component **80** through the filtering component **50**. According to a capacitor calculation formula, $C = \epsilon S / 4 \pi d$, where d is a distance between the human body (head or hands) and the radiator. Therefore, an increase of the capacitance, that is, the sensing signal detected by the detecting component **80** becomes stronger, indicates that the human body is approaching, and a decrease of the capacitance, that is, the sense signal detected by the detecting component **80** becomes weaker, indicates that the human body is moving away. The detecting component **80** determines whether the human's head is close to the sharing radiator **32** of the antenna assembly **100** by detecting changes in the sensing signal mentioned above, so as to intelligently reduce the SAR of the electromagnetic wave by the human's head.

(259) Optionally, at least a part of the DC blocking component **70** can also be a part of the

matching module. For example, the second sub-isolator **72** is a capacitor, the second sub-isolator **72** is configured to block the sensing signal and allow the RF signal to pass, and the second sub-isolator **72** can also be used as a part of the first matching module **M1** to tune impedance match between a signal source and the first feeding point **A**, so as to reduce losses of the RF signal fed to the sharing radiator **32** and to improve conversion efficiency of signals received/transmitted by the sharing radiator **32**. The second sub-isolator **72** can also be used to adjust a frequency offset of resonant modes generated by the sharing radiator **32**, so as to realize multi-use of devices, to reduce the number of devices and space occupied by devices, and to improve the integration of devices. (260) When a part of the first matching module **M1** being electrically connected to the first feeding point **A** is a capacitor, the second sub-isolator **72** does not need to be disposed. Similarly, when a part of the fourth matching module **M4** being electrically connected to the second ground end **212** is a capacitor, the first sub-isolator **71** does not need to be disposed. When a part of the fifth matching module **M5** being electrically connected to the third ground end **311** is a capacitor, the third sub-isolator **73** does not need to be disposed. When a part of the third matching module **M3** being electrically connected to the second feeding point **B** is a capacitor, the fourth sub-isolator **74** does not need to be disposed.

(261) In the antenna assembly **100** and the electronic device **1000** provided in the present disclosure, the sharing radiator **32** on the antenna assembly **100** is reused as a sensing electrode for detecting the approach of the subject to-be-detected such as the human body, and the sensing signal and the RF signal are separated by the DC blocking component **70** and the filtering component **50**. In this way, the dual functions of the antenna assembly **100**, namely, communication and sensing the subject to-be-detected, can be achieved, the function of the antenna assembly **100** is increased, the utilization rate of devices is further improved, and the overall volume of the electronic device **1000** is reduced.

(262) The present embodiment, on one hand, can increase the proximity sensing area, and on the other hand, can increase SAR detection range by taking the sharing radiator **32** as the sensing electrode to detect the approach of the subject to-be-detected at the front and back of the electronic device **1000**, a facing side of the first side frame **2103**, and a facing side of the top frame **2101**, because the sharing radiator **32** is disposed on the first side frame **2103**, the top frame **2101**, and the corner portion **2106** of the first side frame **2103** and the top frame **2101**.

(263) In a second type of embodiment of proximity detecting, as illustrated in FIG. **48**, the second radiator **21** is used as the sensing electrode for detecting the approach of the subject to-be-detected.

(264) The DC blocking component **70** is electrically connected between the second ground end **212** and the second reference ground **GND2** of the second radiator **21**, and between the second matching module **M2** and the second feeding point **B**. For a specific structure of the DC blocking component **70** and principles of isolating sensing signals and allowing RF signals to pass, reference may be made to the connection manner of the first type of DC blocking component **70** and the filtering component **50**, and details are not repeatedly described herein. One end of the filtering component **50** is electrically connected to one end of the DC blocking component **70** close to the second radiator **21** (for example, between the DC blocking component **70** and the second ground end **212**) or electrically connected to any position of the second radiator **21**. The DC blocking component **70** is configured to block direct current generated by the second reference ground **GND2** and the second matching module **M2**. The filtering component **50** is configured to block the RF signal received/transmitted by the second radiator **21** and a sensing signal generated by the second radiator **21** in response to the approach of the subject to-be-detected. In this embodiment, the second radiator **21** is the sensing electrode, and the second radiator **21** is in a “floating state” relative to direct current.

(265) In a third type of embodiment of proximity detecting, as illustrated in FIG. **49**, the fourth radiator **41** is used as the sensing electrode for detecting the approach of the subject to-be-detected. The DC blocking component **70** is electrically connected between the fourth ground end **412** and

the sixth reference ground GND6, and between the sixth matching module M6 and the fourth feeding point D. One end of the filtering component 50 is electrically connected to one end of the DC blocking component 70 close to the third radiator 31 or is electrically connected to the third radiator 31. The DC blocking component 70 is configured to block direct current generated by the sixth reference ground GND6 and the sixth matching module M6. The filtering component 50 is configured to block the RF signal received/transmitted by the third radiator 31 and a sensing signal generated by the fourth radiator 41 in response to the approach of the subject to-be-detected.

(266) It can be understood that, the first type of embodiment of proximity detecting, the second type of embodiment of proximity detecting, and the third type of embodiment of proximity detecting may be implemented by selecting one of them, two of them, or three of them.

(267) The following specifically describes an embodiment in which the first type of embodiment of proximity detecting and the third type of embodiment of proximity detecting are implemented simultaneously.

(268) As illustrated in FIG. 50, based on the first type of embodiment of proximity detecting, the DC blocking component 70 further includes a fifth sub-isolator 75 and a sixth sub-isolator 76. The fifth sub-isolator 75 is electrically connected between the fourth feeding point D and the sixth matching module M6, and the sixth sub-isolator 76 is electrically connected between the fourth ground end 412 and the sixth reference ground GND6, so that both the sharing radiator 32 and the fourth radiator 41 can be used as the sensing electrodes for detecting the approach of the subject to-be-detected.

(269) As illustrated in FIG. 50, the filtering component 50 includes a first sub-filter 51 and a second sub-filter 52. The first sub-filter 51 is electrically connected between the first sub-isolator 71 and the first ground end 111, between the second sub-isolator 72 and the first feeding point A, between the third sub-isolator 73 and the third ground end 311, between the fourth sub-isolator 74 and the third feeding point C, or at any position of the sharing radiator 32. The second sub-filter 52 is electrically connected to any position between the fifth sub-isolator 75 and the fourth feeding point D, between the sixth sub-isolator 76 and the fourth ground end 412, or at any position of the fourth radiator 41.

(270) Specifically, each of the first sub-isolator 71, the second sub-isolator 72, and the third sub-isolator 73 is a blocking capacitor, and each of the first sub-filter 51 and the second sub-filter 52 is a blocking inductor.

(271) As illustrated in FIG. 50, the detecting component 80 is electrically connected to the first sub-filter 51 and the second sub-filter 52. Specifically, one channel of the detecting component 80 is electrically connected to the first sub-filter 51, and the other channel of the detecting component 80 is electrically connected to the second sub-filter 52. In this embodiment, both the sharing radiator 32 and the fourth radiator 41 can be used as detecting electrodes for detecting the approach of the subject to-be-detected.

(272) In other embodiments, the detecting component 80 includes a first sub-detector and a second sub-detector. The first sub-detector is electrically connected to the other end of the first sub-filter 51, and the second sub-detector is electrically connected to the other end of the second sub-filter 52. In other words, sensing signals detected by the sharing radiator 32 are detected by the first sub-detector, and sensing signals detected by the fourth radiator 41 are detected by the second sub-detector. This embodiment may be used when the sharing radiator 32 and the fourth radiator 41 are separately disposed on different sides of the electronic device 1000, and one radiator of the antenna assembly 100 may be used to detect the approach of the human body from different sides of the electronic device 1000, so that the electronic device 1000 can have a larger detecting range while occupying small space.

(273) When the human body approaches the sharing radiator 32, the electrical charge on the sharing radiator 32 changes, and the detecting component 80 may directly detect the sensing signal by the first sub-filter 51. When the human body approaches the fourth radiator 41, the electric

charge in the fourth radiator **41** changes, and the detecting component **80** may directly detect the sensing signal by the second sub-filter **52**. The detecting component **80** detects whether the human body is approaching by detecting the sensing signal. In this case, all radiators can be used as sensing electrodes, so that a sensing area becomes larger, and the utilization rate of radiators can be improved. Only one detecting component **80** is required, so that the number of devices of the antenna assembly **100** can be reduced and the space can be saved.

(274) The controller is electrically connected to the detecting component **80**. The detecting component **80** is configured to receive the sensing signal, to convert the sensing signal into an electrical signal, and to transmit the electrical signal to the controller. The controller is configured to detect the distance between the subject to-be-detected and the radiator according to the magnitude of the sensing signal, so as to determine whether the human body is close to the radiator, and to adjust power of the first feeding module **13** (or the third feeding module **33**, or the fourth feeding module **43**) when the distance between the subject to-be-detected and the radiator is less than or equal to a pre-set distance. Specifically, the controller can adjust power of the antenna assembly **100** according to different scenarios, so as to intelligently reduce the SAR of the electromagnetic wave signal by the human body.

(275) For example, when the human's head is close to the radiator of the antenna assembly **100**, the controller may lower the power of the antenna assembly **100** to lower the SAR of the electromagnetic wave radiated by the antenna assembly **100**. When the human's hand blocks the radiator of the antenna assembly **100** in a radiation direction, if the electronic device **1000** is provided with other standby antenna assemblies **100** (namely, an antenna assembly **100** capable of covering the same frequency band), the controller may turn off the blocked antenna assembly **100** and turn on other unblocked antenna assemblies **100**. In this way, when the human's hand blocks the antenna assembly **100**, the communication quality of the electronic device **1000** can be ensured by intelligently switching the antenna assembly **100**. If the electronic device **1000** is not provided with other standby antenna assemblies **100**, the controller may increase the power of the antenna assembly **100** to compensate for a reduction in efficiency caused by the human's hand blocking the radiator.

(276) Certainly, the controller is also configured to control other application programs on the electronic device **1000** according to a detecting result of the detecting component **80**. For example, if the controller detects that the human body is approaching or the electronic device **1000** is in a call state according to the detection result of the detecting component **80**, the controller can turn off the display screen **300** to save electric energy of the electronic device **1000** during a call; if the controller detects that the human body is moving away or the electronic device **1000** is in a call state, the controller can also turn on the display screen **300**.

(277) Certainly, in other embodiments, the first type of embodiment of proximity detecting embodiment, the second type of embodiment of proximity detecting embodiment, and the third type of embodiment of proximity detecting embodiment may be implemented together.

(278) Optionally, when the radiator is used for detecting the approach of the human body, and the radiator is integrated with the frame **210**, an insulating film may be disposed on a surface of the radiator. Because the human skin surface has an electrical charge, a capacitor structure is formed between the human skin surface and the radiator, and a signal change caused by approach of the human skin surface is detected by the radiator.

(279) Optionally, the electronic device **1000** may be provided with multiple antenna assemblies **100**, and each of the multiple antenna assemblies **100** may be the antenna assembly **100** listed in any one of foregoing embodiments. The multiple antenna assemblies **100** can realize intelligent switching in different blocking scenarios, so as to achieve high antenna receiving/transmitting efficiency in any case of hand blocking, or the like. In addition, after radiators of the multiple antenna assemblies **100** are integrated with a detecting function, different user scenarios (for example, holding with one hand, holding with two hands, a carrying state, a call state, or the like)

may be detected, and six faces in the SAR standard can also be detected.

(280) Optionally, as illustrated in FIG. 51, a segment between the first ground end **111** and the third ground end **311** is a connecting segment. The connecting segment has at least one connecting point (for example, the first connecting point **511** in FIG. 51 and the second connecting point **512** in FIG. 52). The antenna assembly **100** further includes at least one seventh matching module M7. One end of the at least one seventh matching module M7 is electrically connected to the connecting point (namely, the first connecting point **511**), and the other end of the at least one seventh matching module M7 is electrically connected to the seventh reference ground GND7. The at least one seventh matching module M7 can effectively filter out the RF signal of the first antenna unit **10** and the second antenna unit **20**, so as to prevent the RF signal from interfering the third antenna unit **30** and the fourth antenna unit **40**. Optionally, the at least one seventh matching module M7 includes a ground circuit that is in a band-pass or low-impedance state on frequency bands supported by the first antenna unit **10** and the second antenna unit **20**, so that the first antenna unit **10** and the second antenna unit **20** have a better isolation with the third antenna unit **30** and the fourth antenna unit **40**. In addition, the at least one seventh matching module M7 can also effectively filter out the RF signal of the third antenna unit **30** and the fourth antenna unit **40**, so as to prevent the RF signal from interfering the first antenna unit **10** and the second antenna unit **20**. Optionally, the at least one seventh matching module M7 includes the ground circuit which is in a band-pass or low impedance state on frequency bands supported by the third antenna unit **30** and the fourth antenna unit **40**, so that the first antenna unit **10** and the second antenna unit **20** have better isolation with the third antenna unit **30** and the fourth antenna unit **40**.

(281) Furthermore, the at least one seventh matching module M7 can also filter out some working modes with low efficiency in the antenna assembly **100**, for example, a working mode where current flows from the first reference ground GND1 to the first grounding terminal **111**, to the third grounding terminal **311**, and back to the fifth reference ground GND5, so as to reduce influence of low-efficiency working modes on resonant modes of the antenna assembly **100**.

(282) Optionally, the connecting point (namely, the first connecting point **511**) may be close to the first ground end **111** or the third ground end **311**, so that a key can be further disposed between the first ground end **111** and the third ground end **311**. The at least one seventh matching module M7 is opposite to the connecting point. Certainly, when a key is not needed between the first ground end **111** and the third ground end **311**, the connecting point may also be disposed at any position between the first ground end **111** and the third ground end **311**, including but not limited to the middle between the first ground end **111** and the third ground end **311**.

(283) Optionally, as illustrated in FIG. 52, the connecting segment may have multiple connecting points, for example, two connecting points, namely, the first connecting point **511** and the second connecting point **512**, where the first connecting point **511** is close to the first ground end **111**, and the second connecting point **512** is close to the third ground end **311**. Two connecting points (namely, the first connecting point **511** and the second connection **512**) are electrically connected to two matching modules, where one matching module being electrically connected to the first connecting point **511** is marked as the seventh matching module M7, and the other matching module being electrically connected to the second connection **512** is marked as an eighth matching module M8. Each of the seventh matching module M7 and the eighth matching module M8 can make the first antenna unit **10** and the second antenna unit **20** have increasing isolation with the third antenna unit **30** and the fourth antenna unit **40**, and filter some low-efficiency working modes in the antenna assembly **100**, so as to improve the receiving/transmitting efficiency of each resonant mode of the antenna assembly **100**.

(284) The antenna assembly **100** provided in the present disclosure can realize a four-antenna co-aperture design, so that bandwidth of frequency bands received/transmitted and the number of frequency bands received/transmitted can be increased, and overlapping space of the antenna assembly **100** can also be reduced. In addition, the human-body-detecting function is integrated

into radiators of the antenna assembly **100**, so as to intelligently detect whether the human body is approaching, and to reduce the SAR, or to detect the state in which the electronic device **1000** is held and to detect the state of a using application. Furthermore, the antenna assembly **100** is properly disposed on the electronic device **1000**, so that the electronic device **1000** has higher signal receiving/transmitting efficiency in landscape mode.

(285) The above are merely some embodiments of the present disclosure. It should be noted that, those skilled in the art may make further improvements and modifications within the principle of the present disclosure, and these improvements and modifications shall also be included in the scope of protection of the present disclosure.

Claims

1. An antenna assembly, comprising: a first radiator, a second radiator, a first matching module, a first feeding module, a second matching module, and a second feeding module, wherein the first radiator has a first ground end, a first coupling end, and a first feeding point between the first ground end and the first coupling end; the second radiator has a second coupling end, a second ground end, and a second feeding point between the second coupling end and the second ground end; a first coupling gap is defined between the second coupling end and the first coupling end; the first matching module is electrically connected between the first feeding point and the first feeding module, and the first ground end is electrically connected to a first reference ground; the second matching module is electrically connected between the second feeding point and the second feeding module, and the second ground end is electrically connected to a second reference ground; and the first radiator and the second radiator support, under excitation of the first feeding module and the second feeding module, a plurality of resonant modes, wherein the plurality of resonant modes comprise a first resonant mode, a second resonant mode, and a third resonant mode, the first resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the first feeding module resonating on the first radiator, the second resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by excitation current of the first feeding module resonating on the second radiator, and the third resonant mode is a $\frac{3}{4}$ wavelength mode caused by the excitation current of the second feeding module resonating on the first radiator.
2. The antenna assembly of claim 1, wherein the plurality of resonant modes further comprise a fourth resonant mode, and the fourth resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength caused by the excitation current of the first feeding module resonating from the first feeding point to the first coupling end.
3. The antenna assembly of claim 2, wherein a resonant frequency of the first resonant mode, a resonant frequency of the second resonant mode, and a resonant frequency of the fourth resonant mode increase sequentially; the first resonant mode supports at least one of a global positioning system (GPS) frequency band, a long-term evolution 4th generation (LTE-4G) middle-high band (MHB) frequency band, or a new radio 5th generation (NR-5G) MHB frequency band; the second resonant mode supports at least one of a wireless fidelity 2.4 GHz (Wi-Fi 2.4G) frequency band, the LTE-4G MHB frequency band, or the NR-5G MHB frequency band; and the fourth resonant mode supports at least one of an LTE-4G ultra-high band (UHB) frequency band or an NR-5G UHB frequency band.
4. The antenna assembly of claim 2, wherein the first matching module comprises at least one first branch connected in parallel to the ground and/or at least one second branch connected in series between the first feeding point and the first feeding module, and each of the at least one first branch and the at least one second branch comprises at least one of a capacitor or an inductor; and the first matching module is configured to tune at least one of a resonant frequency of the first resonant mode, a resonant frequency of the second resonant mode, or a resonant frequency of the fourth resonant mode.

5. The antenna assembly of claim 4, wherein the at least one first branch comprises a first sub-branch, a second sub-branch, and a third sub-branch, and the at least one second branch comprises a fourth sub-branch and a fifth sub-branch, wherein: one end of the first sub-branch is electrically connected to the first feeding point, another end of the first sub-branch is electrically connected to a third reference ground, and the first sub-branch comprises a capacitor, or a capacitor and an inductor; one end of the second sub-branch is electrically connected to the first feeding point, another end of the second sub-branch is electrically connected to the third reference ground, and the second sub-branch comprises an inductor, or an inductor and a capacitor; another end of the third sub-branch is electrically connected to the third reference ground, and the third sub-branch comprises a capacitor, or a capacitor and an inductor; one end of the fourth sub-branch is electrically connected to the first feeding point, another end of the fourth sub-branch is electrically connected to one end of the third sub-branch, and the fourth sub-branch comprises a capacitor, or a capacitor and an inductor; and one end of the fifth sub-branch is electrically connected to said another end of the fourth sub-branch, another end of the fifth sub-branch is electrically connected to the first feeding module, and the fifth sub-branch comprises an inductor.
6. The antenna assembly of claim 5, wherein the first matching module further comprises at least one adjustable branch electrically connected to the third reference ground, the at least one adjustable branch each comprises at least one of a switching circuit or an adjustable capacitor, and the at least one adjustable branch is configured to tune the resonant frequency of the first resonant mode and the resonant frequency of the second resonant mode.
7. The antenna assembly of claim 2, wherein the plurality of resonant modes further comprise a fifth resonant mode, and the fifth resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by excitation current of the second feeding module resonating from the second feeding point to the second coupling end.
8. The antenna assembly of claim 7, wherein a resonant frequency of the fifth resonant mode is less than a resonant frequency of the third resonant mode, and the fifth resonant mode supports at least one of a Wi-Fi 5 GHz (Wi-Fi 5G) frequency band or a Wi-Fi 6 GHz extended (Wi-Fi 6E) frequency band, and the third resonant mode supports at least one of a Wi-Fi 5G frequency band or a Wi-Fi 6E frequency band.
9. The antenna assembly of claim 7, wherein the second matching module comprises at least one third branch connected in parallel to the ground and/or at least one fourth branch connected in series between the first feeding point and the first feeding module, and each of the at least one third branch and the at least one fourth branch comprises at least one of a capacitor or an inductor; and the second matching module is configured to allow a frequency band supported by the fifth resonant mode and a frequency band supported by the third resonant mode to pass, to block a frequency band less than both the frequency band supported by the fifth resonant mode and the frequency band supported by the third resonant mode, and to tune a resonant frequency of the fifth resonant mode and/or a resonant frequency of the third resonant mode.
10. The antenna assembly of claim 9, wherein the at least one third branch comprises a sixth sub-branch and a seventh sub-branch, and the at least one fourth branch comprises an eighth sub-branch; wherein one end of the sixth sub-branch is electrically connected to the second feeding point, another end of the sixth sub-branch is electrically connected to one end of the seventh sub-branch, and the sixth sub-branch comprises an inductor and a capacitor connected in parallel; wherein another end of the seventh sub-branch is electrically connected to the second feeding module, and the seventh sub-branch comprises a capacitor, or an inductor, or an inductor and a capacitor connected in parallel; and wherein one end of the eighth sub-branch is electrically connected to the second feeding module, another end of the eighth sub-branch is electrically connected to a fourth reference ground, and the eighth sub-branch comprises at least one of a capacitor or an inductor.
11. The antenna assembly of claim 1, further comprising a third radiator, a third feeding module,

and a third matching module, wherein the third radiator has a third ground end, a first free end, and a third feeding point between the third ground end and the first free end, the third ground end and the first ground end are spaced apart from each other or are connected via a conductor, and the third ground end is electrically connected to a fifth reference ground; and the third feeding module is electrically connected to the third feeding point, and the third matching module is electrically connected between the third feeding point and the third feeding module.

12. The antenna assembly of claim 11, further comprising: a fourth matching module, wherein one end of the fourth matching module is electrically connected to the first ground end, and another end of the fourth matching module is electrically connected to the first reference ground; and/or a fifth matching module, wherein one end of the fifth matching module is electrically connected to the third ground end, and another end of the fifth matching module is electrically connected to the fifth reference ground.

13. The antenna assembly of claim 11, further comprising a fourth radiator, a fourth feeding module, and a sixth matching module, wherein the fourth radiator has a fourth ground end, a second free end, and a fourth feeding point between the fourth ground end and the second free end, a second coupling gap is defined between the fourth ground end and the first free end, and the fourth ground end is electrically connected to a sixth reference ground; and the fourth feeding module is electrically connected to the fourth feeding point, and the sixth matching module is electrically connected between the fourth feeding point and the fourth feeding module.

14. The antenna assembly of claim 13, wherein: the antenna assembly is further configured to support a sixth resonant mode, a seventh resonant mode, and an eighth resonant mode, wherein the sixth resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by excitation current of the third feeding module resonating on the third radiator; the seventh resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module resonating from the third feeding point to the first free end; and the eighth resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module resonating from the third feeding point to the first free end and is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the third feeding module resonates from the second free end to the fourth ground end; and the antenna assembly is further configured to support a ninth resonant mode, a tenth resonant mode, an eleventh resonant mode, and a twelfth resonant mode, wherein the ninth resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by excitation current of the fourth feeding module resonating from the third feeding point to the first free end and is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the fourth feeding module resonating from the first free end to the fourth ground end; the tenth resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the fourth feeding module resonating from the first free end to the fourth ground end; the eleventh resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the fourth feeding module resonating from the fourth feeding point to the second free end; and the twelfth resonant mode is a $\frac{3}{4}$ wavelength mode caused by the excitation current of the fourth feeding module resonating from the third ground end to the first free end.

15. The antenna assembly of claim 13, wherein the third radiator supports, under excitation of the third feeding module, a GPS frequency band, a Wi-Fi 2.4G frequency band, an LTE-4G MHB frequency band, an NR-5G MHB frequency band, an LTE-4G UHB frequency band, and an NR-5G UHB frequency band; and the fourth radiator supports, under excitation of the fourth feeding module, an N77 frequency band, an N78 frequency band, a Wi-Fi 5G frequency band, and a Wi-Fi 6E frequency band.

16. The antenna assembly of claim 13, wherein the first radiator is connected to the third radiator to form a sharing radiator, and the antenna assembly further comprises a direct-current (DC) blocking component, a filtering component, and a detecting component, wherein the DC blocking component is electrically connected between the first feeding point and the first matching module, between the first ground end and the first reference ground, between the third ground end and the

fifth reference ground, and between the third feeding point and the third matching module; one end of the filtering component is electrically connected to one end of the DC blocking component close to the sharing radiator or is electrically connected to the sharing radiator; and the DC blocking component is configured to block direct current generated by the first matching module, the first reference ground, the fifth reference ground, and the third matching module, and the filtering component is configured to block a radio frequency (RF) signal received/transmitted by the sharing radiator and to block a sensing signal generated by the sharing radiator in response to approach of a subject to-be-detected; and/or wherein the DC blocking component is electrically connected between the second ground end and the second reference ground, and between the second matching module and the second feeding point; one end of the filtering component is electrically connected to one end of the DC blocking component close to the second radiator or is electrically connected to the second radiator; and the DC blocking component is configured to block direct current generated by the second reference ground and the second matching module, and the filtering component is configured to block an RF signal received/transmitted by the second radiator and to block a sensing signal generated by the second radiator in response to approach of the subject to-be-detected; and/or wherein the DC blocking component is electrically connected between the fourth ground end and the sixth reference ground, and between the sixth matching module and the fourth feeding point; one end of the filtering component is electrically connected to one end of the DC blocking component close to the third radiator or is electrically connected to the third radiator; and the DC blocking component is configured to block direct current generated by the sixth reference ground and the sixth matching module, and the filtering component is configured to block an RF signal received/transmitted by the third radiator and to block a sensing signal generated by the third radiator in response to approach of the subject to-be-detected; and wherein the detecting component is electrically connected to another end of the filtering component, and the detecting component is configured to detect magnitude of the sensing signal.

17. The antenna assembly of claim 13, wherein a segment between the first ground end and the third ground end is a connecting segment, the antenna assembly further comprises at least one seventh matching module, one end of the at least one seventh matching module is electrically connected to the connecting segment, and another end of the seventh matching module is connected to the ground.

18. An electronic device, comprising: a housing and an antenna assembly, wherein radiators of the antenna assembly are integrated into the housing, or are disposed on a surface of the housing, or are disposed in space enclosed by the housing; wherein the antenna assembly comprises a first radiator, a second radiator, a first matching module, a first feeding module, a second matching module, and a second feeding module, wherein the first radiator has a first ground end, a first coupling end, and a first feeding point between the first ground end and the first coupling end; the second radiator has a second coupling end, a second ground end, and a second feeding point between the second coupling end and the second ground end; a first coupling gap is defined between the second coupling end and the first coupling end; the first matching module is electrically connected between the first feeding point and the first feeding module, and the first ground end is electrically connected to a first reference ground; the second matching module is electrically connected between the second feeding point and the second feeding module, and the second ground end is electrically connected to a second reference ground; and wherein the first radiator and the second radiator support, under excitation of the first feeding module and the second feeding module, a plurality of resonant modes, wherein the plurality of resonant modes comprise a first resonant mode, a second resonant mode, and a third resonant mode, the first resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by the excitation current of the first feeding module resonating on the first radiator, the second resonant mode is a $\frac{1}{8}$ to $\frac{1}{4}$ wavelength mode caused by excitation current of the first feeding module resonating on the second radiator, and the third resonant mode is a $\frac{3}{4}$ wavelength mode caused by the excitation current of the second feeding module resonating on the

first radiator.

19. The electronic device of claim 18, wherein the housing has a top frame and a bottom frame disposed opposite to the top frame and a first side frame and a second side frame connected between the top frame and the bottom frame, and the antenna assembly is disposed on at least one of the top frame, the first side frame, the second side frame, or the bottom frame.

20. The electronic device of claim 19, wherein a first radiator of the antenna assembly and a second radiator of the antenna assembly are disposed on the first side frame, one part of a third radiator of the antenna assembly is disposed on the first side frame, and another part of the third radiator of the antenna assembly is disposed on the top frame, a fourth radiator of the antenna assembly is disposed on the top frame, a first coupling gap between the first radiator and the second radiator is defined on the first side frame, and a second coupling gap is defined on the top frame; and wherein the electronic device further comprises at least one key portion, the at least one key portion is disposed between a second ground end and a second feeding point, and/or the at least one key portion is disposed between a first ground end and a third ground end of the third radiator.
